

True pelvis, pelvic floor and perineum

TRUE PELVIS AND PELVIC FLOOR

The true pelvis is a bowl-shaped structure formed from the sacrum, pubis, ilium, ischium, the ligaments that interconnect these bones, and the muscles that line their inner surfaces. The true pelvis is considered to start at the level of the plane passing through the promontory of the sacrum, the arcuate line on the ilium, the iliopectineal line and the posterior surface of the pubic crest. This plane, or 'inlet', lies at an angle of between 35° and 50° up from the horizontal; above this plane, the bony structures are sometimes referred to as the false pelvis. They form part of the walls of the lower abdomen. In children, the width of the pelvic inlet is an age-independent predictor of chest width and thoracic dimensions (Emans et al 2005). The 'outlet' of the true pelvis is formed by the ischiopubic rami, ischial tuberosities, sacrotuberous ligaments and distal sacrum. The bones surround a central pelvic canal that forms a ventrally concave curve (the curve of Carus); in the female, it constitutes the birth canal. The details of the topography of the bony and ligamentous pelvis are considered fully in Chapter 80.

MUSCLES AND FASCIAE OF THE PELVIS

Pelvic muscles

The muscles arising within the pelvis form two groups. Piriformis and obturator internus form part of the walls of the pelvis, and are considered primarily as muscles of the lower limb (Fig. 73.1). Levator ani and

ischiococcygeus form the pelvic diaphragm and delineate the lower limit of the true pelvis (Fig. 73.2). The fasciae investing the muscles are continuous with visceral pelvic fascia above, perineal fascia below, and obturator fascia laterally.

Piriformis

Piriformis forms part of the posterolateral wall of the true pelvis and is attached to the anterior surface of the sacrum, the gluteal surface of the ilium near the posterior inferior iliac spine, the capsule of the adjacent sacroiliac joint and, sometimes, to the upper part of the pelvic surface of the sacrotuberous ligament. It passes out of the pelvis through the greater sciatic foramen above the sacrospinous ligament. Within the pelvis, the posterior surface of the muscle lies against the sacrum, and the anterior surface is related to the rectum (especially on the left), the sacral plexus of nerves and branches of the internal iliac vessels. Piriformis is described in more detail in Chapter 80.

Obturator internus

Obturator internus and the fascia over its upper, inner (pelvic), surface form part of the anterolateral wall of the true pelvis. It is attached to the structures surrounding the obturator foramen, ischio-pubic ramus, the pelvic surface of the hip bone below and behind the pelvic brim, and the upper part of the greater sciatic foramen. It is also attached to the medial part of the pelvic surface of the obturator membrane. The muscle is covered by a fascial layer, and the muscle fibres can be seen through this semi-transparent membrane from within the pelvis. Specialized portions of the fascia give attachment to some of the fibres of levator ani (tendinous arch of levator ani), so that only the upper

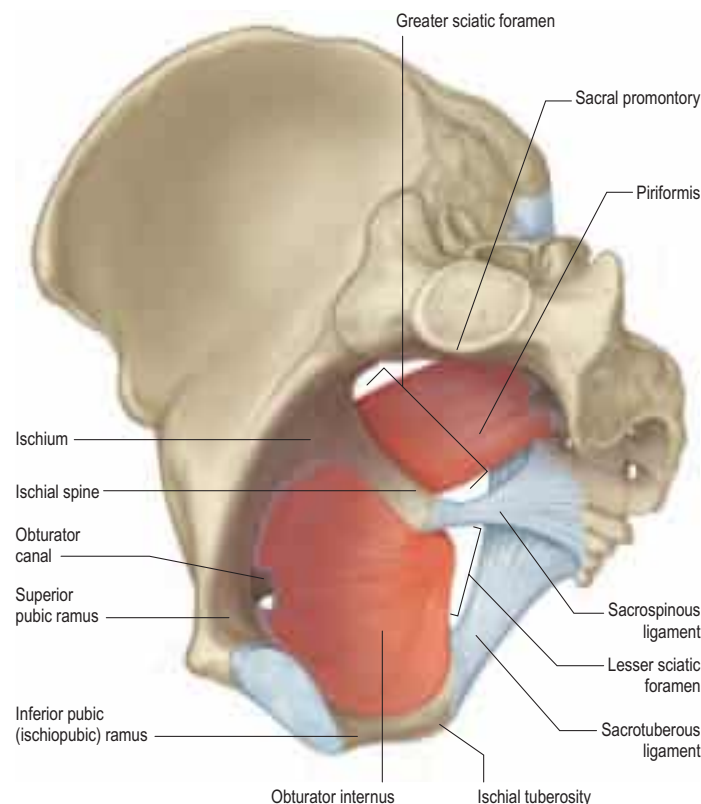


Fig. 73.1 Piriformis, obturator internus and the ligaments of the pelvis. Those muscles relating only to the pelvis or perineum have been omitted for clarity. (Adapted with permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

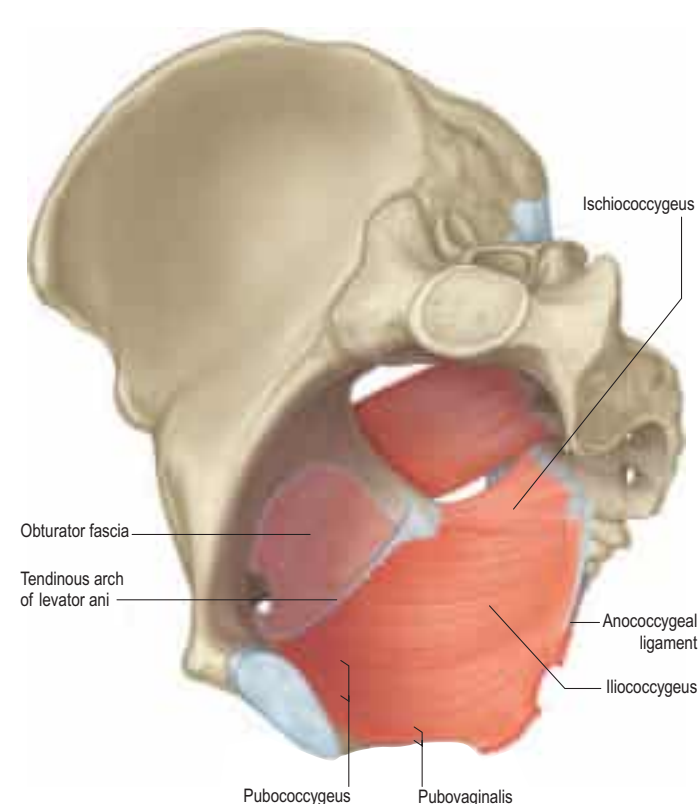


Fig. 73.2 Muscles of the female pelvis. The superior gluteal and obturator vessels and nerves, as well as the pelvic viscera, have been omitted for clarity. (Adapted with permission from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

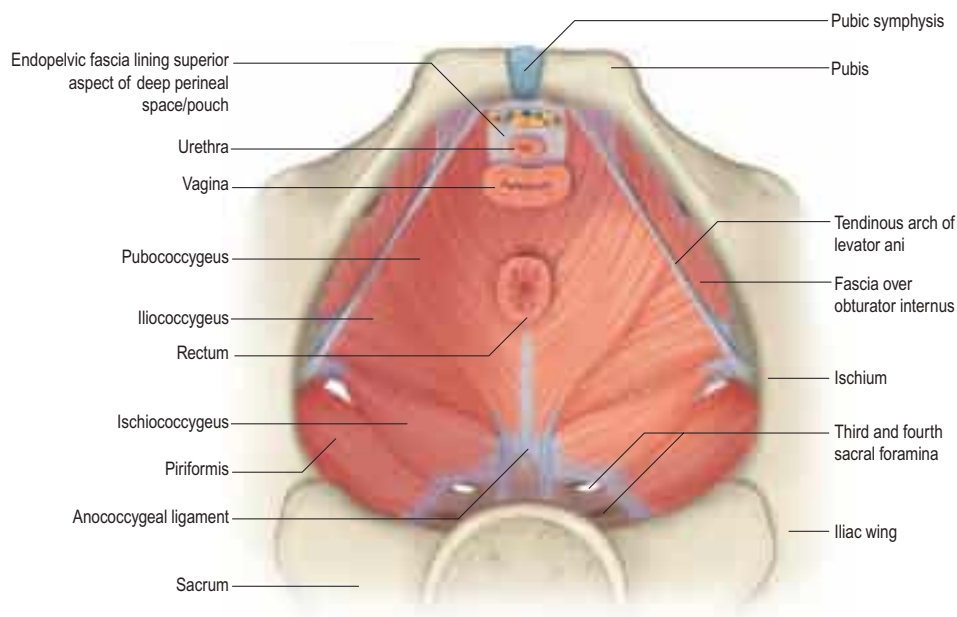


Fig. 73.3 Muscles of the female pelvis viewed from above. The sacral nerve roots have been divided close to the sacral foramina. The anorectal junction, vagina and urethra have been divided at the level of the pelvic floor.

portion of obturator internus can be seen from above. The lower portion forms part of the boundaries of the ischio-anal fossa. In the male, the upper portion lies lateral to the bladder, the obturator and vesical vessels, and the obturator nerve. In the female, the attachments of the broad ligament of the uterus, the ovarian end of the uterine tubes and the uterine vessels also lie medial to obturator internus and its fascia.

Levator ani (pubococcygeus, iliococcygeus and puborectalis)

Levator ani is a broad muscular sheet of variable thickness attached to the internal surface of the pelvis. It forms a large portion of the pelvic floor (**Fig. 73.3**) ([Lawson 1974a](#), [Roberts et al 1988](#), [Wendell-Smith and Wilson 1991](#)). The muscle is subdivided into named portions according to their attachments and the pelvic viscera to which they are related (pubococcygeus, iliococcygeus and puborectalis). These parts are often referred to as separate muscles but the boundaries between each part cannot be easily distinguished and, moreover, they perform many similar physiological functions. Ischiococcygeus (coccygeus) lies immediately cranial to levator ani and is contiguous with it. Pubococcygeus is often subdivided into separate parts according to the pelvic viscera to which each part relates (puboperinealis, puboprostaticus or pubovaginalis, puboanalis, puborectalis). Levator ani arises from each side of the walls of the pelvis along the condensation of the obturator fascia (the tendinous arch of levator ani). Fibres from ischiococcygeus attach to the sacrum and coccyx but the remaining parts of the muscle converge in the midline. Fibres from iliococcygeus join by a partly fibrous intersection and form the iliococcygeal raphe posterior to the anorectal junction. Closer to the anorectal junction, and elsewhere in the pelvic floor, the fibres are more nearly continuous with those of the opposite side, such that the muscle forms a sling (iliococcygeus, puborectalis).

Attachments The attachments for pubococcygeus, iliococcygeus and puborectalis are as follows.

Pubococcygeus Pubococcygeus originates from the posterior aspect of the body of the pubis and passes back almost horizontally. The most medial fibres run directly lateral to the urethra and its sphincter as it passes through the pelvic floor; here, the muscle is correctly called puboperinealis, although, because of its close relationship to the upper half of the urethra in both sexes, it is often referred to as pubourethralis; despite this, no direct connection with the urethra is present. In males, some of these fibres lie lateral and inferior to the prostate and are referred to as puboprostaticus (levator prostatae). In females, fibres run further back and attach to the lateral walls of the vagina, where they are referred to as pubovaginalis. In both sexes, fibres from this part of pubococcygeus attach to the perineal body; a few elements also attach to the anorectal junction. Some fibres, sometimes called puboanalis, decussate and blend with the longitudinal rectal muscle and fascial elements to contribute to the conjoint longitudinal coat of the anal canal. Behind the rectum, other fibres of pubococcygeus form a tendinous intersection as part of the levator raphe. Elements of pubococcygeus (puboperineus, pubovaginalis/puboprostaticus and puboanalis) may collectively be referred to as 'pubovisceralis'.

Iliococcygeus Iliococcygeus is attached to the inner surface of the ischial spine below and anterior to the attachment of ischiococcygeus, and also to the tendinous arch as far forwards as the obturator canal (see **Fig. 73.2**). The most posterior fibres are attached to the tip of the sacrum and coccyx, but the majority join fibres from the contralateral side to form a raphe that is effectively continuous with the fibroelastic anococcygeal ligament, which is closely applied to its inferior surface; some muscle fibres may attach into the ligament. The raphe provides a strong attachment for the pelvic floor posteriorly and must be divided to allow wide excisions of the anorectal canal during abdominoperineal excisions for malignancy. An accessory slip may arise from the most posterior part and is sometimes referred to as iliosacralis.

Puborectalis Puborectalis lies lateral to pubococcygeus and cannot be seen from inside the pelvis. It originates from the inner surface of the ischiopubic rami immediately adjacent to, and sometimes arising in part from, the perineal membrane. Its fibres pass lateral to those of iliococcygeus and pubococcygeus to decussate posterior to the rectum at the anorectal junction. The border between puborectalis and some fibres of the external anal sphincter is indistinct.

Ischiococcygeus Ischiococcygeus may be referred to as a separate muscle, sometimes named coccygeus. It lies as the most posterosuperior portion of levator ani, and arises as a triangular musculotendinous sheet with its apex attached to the pelvic surface and tip of the ischial spine, and base attached to the lateral margins of the coccyx and the fifth sacral segment. Ischiococcygeus is rarely absent, but may be almost completely tendinous rather than muscular. It lies on the pelvic aspect of the sacrospinous ligament and may be fused with it, particularly if it is mostly tendinous. The sacrospinous ligament may represent either a degenerate part or an aponeurosis of the muscle, since muscle and ligament are coextensive.

Relations The superior, pelvic, surface of levator ani is separated only by fascia (superior pelvic diaphragmatic, visceral and extraperitoneal) from the urinary bladder, prostate or uterus and vagina, rectum and peritoneum. Its inferior, perineal, surface forms the medial wall of the ischio-anal fossa and the superior wall of the anterior recess of the fossa, both being covered by inferior pelvic diaphragmatic fascia. The posterior border is separated from the coccyx by areolar tissue. The medial borders of the two levator muscles and the inferior ischiopubic rami border the levator hiatus, through which pass the urethra, vagina (in the female) and anus. In the female, that portion of the hiatus that lies anterior to the perineal body is referred to as the urogenital hiatus.

Vascular supply Levator ani is supplied by branches of the inferior gluteal, inferior vesical and pudendal arteries.

Innervation The nerves to levator ani originate mainly from the third and fourth sacral spinal segments, with lesser contributions from the second segment. These nerves enter the pelvis just above, and sometimes pierce, ischiococcygeus to pass along the ventral surface of ischiococcygeus and pubococcygeus, supplying these muscles and sending fibres to puborectalis ([Sato 1980](#), [Roberts and Krishningner 1967](#)). The

puddendal nerve also may supply pubococcygeus from its lateral surface through its inferior rectal and perineal branches.

Actions Pubococcygeus and puborectalis pull the pelvic and perineal structures ventrally and cranially, occluding the levator hiatus. In the female, this action occludes the vaginal canal and prevents prolapse of the pelvic organs through the urogenital hiatus. The constant baseline activity of the levator muscles is similar to that of the anal sphincter, modulated to adjust to the loads placed on them. The action of puborectalis and pubococcygeus also reinforces the external anal sphincter and helps to create the anorectal angle. Iliococcygeus and, to a lesser extent, the less muscular ischiococcygeus form a relatively horizontal diaphragm, especially in the dorsal half of the pelvis, that assists puborectalis in achieving anorectal and urinary continence.

Levator ani must relax appropriately to permit expulsion of urine and, particularly, faeces; it contracts with the abdominal muscles and the abdominotheracic diaphragm to raise intra-abdominal pressure. It forms much of the muscular pelvic diaphragm, which supports the pelvic viscera. Like the abdominotheracic diaphragm, but unlike the abdominal muscles, levator ani is also active in the inspiratory phase of quiet respiration. In the pregnant female, the shape of the pelvic floor may help to direct the fetal head into the anteroposterior diameter of the pelvic outlet.

Pelvic fasciae

The pelvic fasciae may be conveniently divided into the parietal pelvic fascia, which forms the coverings of the pelvic muscles, and the visceral pelvic fascia, which forms the coverings of the pelvic organs and their neurovascular supply (Fig. 73.4).

Parietal pelvic fascia

The parietal pelvic fascia consists of the obturator fascia, the fasciae over piriformis, and over levator ani (the pelvic diaphragm) and the presacral fascia.

Obturator fascia

The parietal pelvic fascia on the pelvic (medial) surface of obturator internus is well differentiated. In humans, ventral to the lateral attachment of the pelvic organs, a portion of it is derived from the degener-

ated upper portion of the attachment of levator ani. It is connected to the posterior part of the arcuate line of the ilium and is continuous with iliac fascia. Anterior to this, as it follows the line of origin of obturator internus, it is gradually separated from the attachment of the iliac fascia, and a portion of the periosteum of the ilium and pubis spans between them. It arches below the obturator vessels and nerve, investing the obturator canal, and is attached anteriorly to the back of the pubis. Behind the obturator canal, the fascia is markedly aponeurotic and gives a firm attachment to the iliococcygeal portion of levator ani, usually called the tendinous arch of levator ani (arcus tendineus musculi levatoris ani) (see Figs 73.3, 73.14, 73.15). Above the attachment of levator ani, the fascia is thin and is effectively composed only of the epimysium of the muscle and overlying connective tissue; posteriorly, it forms part of the lateral wall of the ischio-anal fossa in the perineum, and anteriorly, it merges with the fasciae of the muscles of the deep perineal space, which is continuous with the ischio-anal fossa. The obturator fascia is continuous with the pelvic periosteum and, thus, the fascia over piriformis.

Fascia over piriformis

The fascia over the inner aspect of piriformis is very thin, and fuses with the periosteum on the front of the sacrum at the margins of the anterior sacral foramina. It ensheathes the anterior primary rami of the sacral nerves that emerge from these foramina; the nerves are often described as lying behind the fascia. The internal iliac vessels lie in front of the fascia over piriformis; their branches draw out sheaths of the fascia and extraperitoneal tissue into the gluteal region, above and below piriformis.

Fascia over levator ani (pelvic diaphragm)

Both surfaces of levator ani have a fascial covering; the combination of the two fascial layers and the intervening muscle is called the pelvic diaphragm. On the inferior surface, the thin fascia is continuous with the obturator fascia below the tendinous arch of levator ani laterally. It covers the medial wall of the ischio-anal fossa and blends below with fasciae on the urethral sphincter and the external anal sphincter. The superior fascia of the pelvic diaphragm is markedly thicker than the inferior fascia and is attached anteriorly to the posterior aspect of the body of the pubis, approximately 2 cm above its lower border. It extends laterally across the superior pubic ramus, blending with the

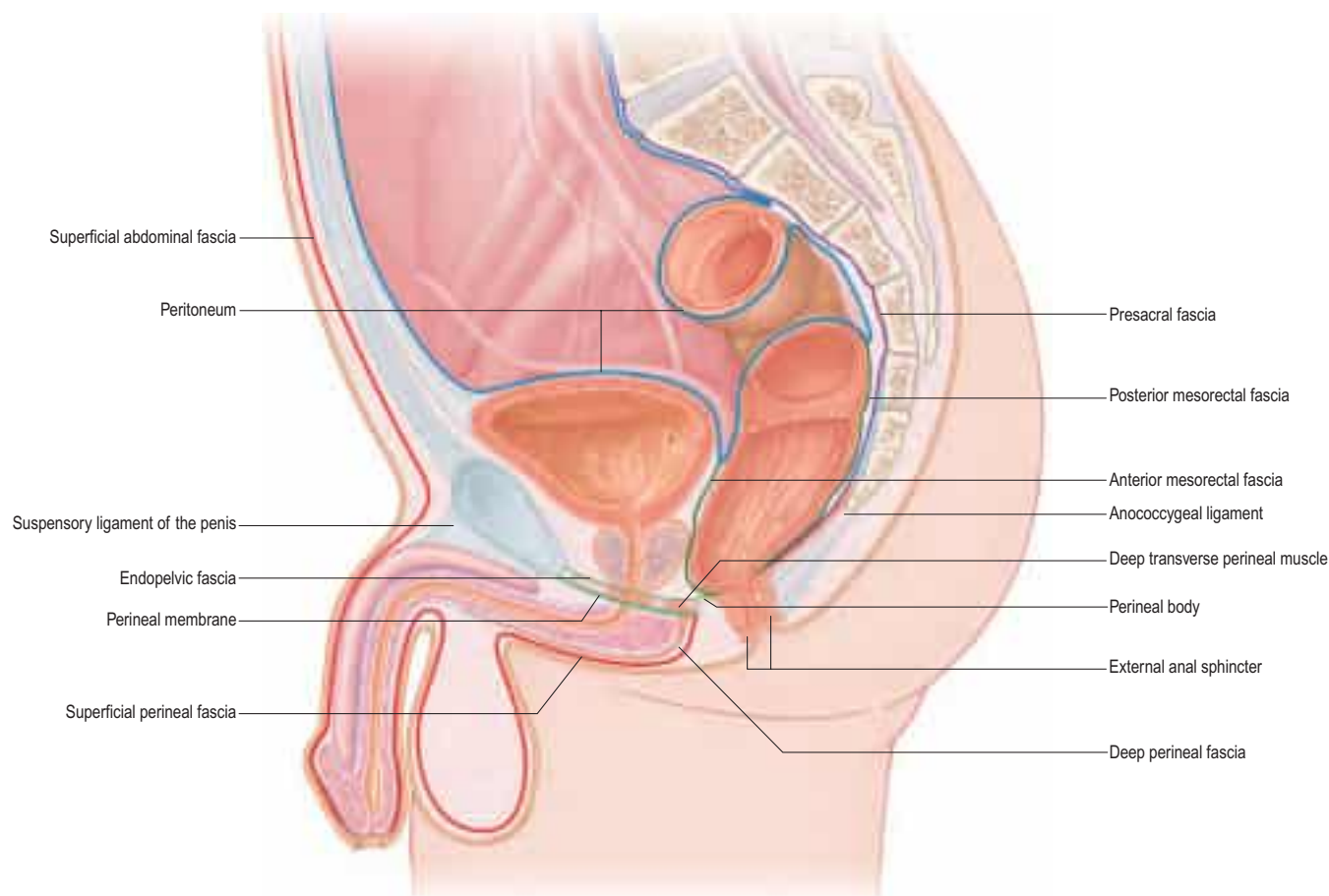


Fig. 73.4 Fasciae of the pelvis and perineum: median sagittal section in the male. The visceral parietal fasciae have been omitted for clarity.

obturator fascia and continuing along an irregular line to the spine of the ischium. It is continuous posteriorly with the fascia over piriformis and the anterior sacrococcygeal ligament. Medially, the superior fascia of the pelvic diaphragm blends with the visceral pelvic fascia to contribute to the endopelvic fascia.

Tendinous arch of the pelvic fascia/white line of the parietal pelvic fascia

Low on the superomedial aspect of the upper fascia over levator ani, a thick, white band of condensed connective tissue extends from the lower part of the pubic symphysis to the superior margin of the ischial spine. It provides attachment for the condensations of visceral pelvic fascia that provide support to the urethra and bladder, and to the vagina in females (see below).

Presacral fascia

The presacral fascia forms a hammock-like structure behind the posterior portion of the mesorectal fascia. Laterally, it extends to the origin of the fascia over piriformis and the fascia over levator ani (superior pelvic diaphragmatic fascia), with which it blends; more inferiorly, it extends between the white line of the parietal pelvic fascia on either side. Inferiorly, it extends to the anorectal junction, where it fuses with the posterior aspect of the mesorectal fascia and the iliococcygeal raphe at the level of the anorectal junction. Superiorly, it can be traced to the origin of the superior hypogastric plexus, where it becomes progressively thinner over the sacral promontory and becomes continuous with the retroperitoneal tissues. The right and left hypogastric nerves and inferior hypogastric plexuses lie on its surface, and the presacral veins lie immediately posterior to it. It forms a distinct layer that can be seen both on magnetic resonance images of the pelvis and during surgery. The presacral fascia provides an important landmark because extension of rectal tumours through it significantly reduces the possibility of curative resectional surgery. Dissection in the plane posterior to the fascia may result in bleeding from the presacral veins; because the adventitia of the veins is partly attached to the posterior surface of the fascia, the haemorrhage may be severe (as the veins are unable to contract properly). The presacral fascia is a useful structure to which the rectum may be sutured during rectopexy for rectal prolapse in children.

Visceral pelvic fascia

The urogenital organs in both sexes are connected bilaterally to the pelvic walls by neurovascular mesenteric condensations ensheathed by a meshwork of loose connective and adipose tissue and lying above the perineal membrane (Roberts et al 1964, Ricci et al 1947, Reiffenstahl 1982, Range and Woodburne 1964, Campbell 1950). The lateral attachments of the pelvic organs to the pelvic side walls are referred to as the endopelvic fascia. Considered as a unit, the connections provide a conduit for conducting neurovascular elements from the pelvic side wall to the organs and attachments that help to retain the pelvic organs in place; this factor is important in the female in preventing pelvic organ prolapse (DeLancey 1992). The loose connective tissue associated with these mesenteries extends to the midline, separating the bladder from the vagina, and the vagina from the rectum, in the female; it separates the bladder, prostate and seminal vesicles from the rectum in the male. The fascial tissues contain varying amounts of connective tissue and smooth muscle; where they either are unusually dense or form visible ridges, they are called 'ligaments' (e.g. cardinal ligament, uterosacral ligament). Clinically, these lateral attachments are often referred to as visceral 'ligaments', but they are mesenteric in nature and, therefore, quite unlike the bands of dense, regular, connective tissue that typify skeletal ligaments. The lateral attachments of the mesenteries sweep off the pelvic walls, arising from the superior fascia over levator ani and from part of the fascia over piriformis more superiorly and posteriorly. The mesenteries passing to the bladder in the male, or the bladder and upper vagina and uterus in the female, are relatively long, but these lateral connections become shorter towards the pelvic outlet, until at the level of the perineal membrane, there is a direct connection between the organs and the pelvic walls.

In the female, the cardinal ligament is the upper portion of this mesentery. It surrounds the cervicovaginal junction and extends down to mid-vagina, where the vagina has a more direct lateral attachment at the tendinous arch of the pelvic fascia. The portions that attach to the uterus and vagina are sometimes called the parametrium and paracolpium, respectively. Further accounts of the paravisceral portions of the visceral pelvic fascia are given in the chapters describing the organs to which the fascia relates.

The uterosacral ligament is a visible fold of tissue flanking the rectum as it descends posterior to the cervix in the female. It contains a considerable amount of smooth muscle near its attachment to the

cervix, together with some of the pelvic autonomic nerves. A similar vesicosacral fold is present in the male.

Approximately 1 cm above the inferior border of the pubic bone and 1 cm lateral to the midline, a band of dense pelvic connective tissue – the anterior end of the tendinous arch of the pelvic fascia – attaches to the paravaginal tissues in the female and the prostatic tissues in the male. This band extends on the inner surface of levator ani and joins the tendinous arch of levator ani to the ischium, just above the spine. The attachment of the paravaginal tissue to the pubic bones is sometimes called the pubourethral ligament, which is a misnomer since it is not attached to the urethra. The attachment of the anterior vaginal wall to the tendinous arch of the pelvic fascia, the paravaginal attachment, helps to provide support to the vagina, urethra and bladder.

There is much less condensation of connective tissue around the rectum. A layer presumed to be a peritoneal fusion fascia is described between the rectum and either the seminal vesicles in the male or the vagina in the female (the rectovesical septum or rectovaginal septum, respectively); it does not connect to the rectum itself. The connective tissue over the longitudinal muscles of the rectum is thickened just above the anal hiatus in levator ani and fuses with the endopelvic fascia and the anococcygeal ligament, forming a structure that is sometimes referred to as a rectosacral ligament.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE OF THE PELVIS

The true pelvis contains the internal iliac arteries and veins, and the lymphatics that drain the majority of the pelvic viscera. The common and external iliac vessels and the lymphatics that drain the lower limb lie along the pelvic brim and in the lower retroperitoneum, but are conveniently discussed together with the vessels of the true pelvis. Remarkable variation exists in the terminal branching pattern for the iliac vessels and no two individuals have quite the same anatomy. General patterns do, however, exist and this description will consider the common pattern.

Arteries of the pelvis

Common iliac arteries

The abdominal aorta bifurcates into the right and left common iliac arteries anterolateral to the left side of the body of the fourth lumbar vertebra (Roberts and Krishningner 1967). These arteries diverge as they descend and they divide at the level of the sacroiliac joint into external and internal iliac arteries. The external iliac artery is the principal artery of the lower limb. The internal iliac artery provides the principal supply to the walls and viscera of the pelvis, the perineum and the gluteal region.

Right common iliac artery The right common iliac artery is approximately 5 cm long. It passes obliquely across part of the bodies of the fourth and the fifth lumbar vertebrae, and is crossed anteriorly by the sympathetic rami to the pelvic plexus and, at its division into internal and external iliac arteries, by the ureter. It is covered by the parietal peritoneum, which separates it from the coils of the small intestine. Posteriorly, it is separated from the bodies of the fourth and fifth lumbar vertebrae and their intervening disc by the right sympathetic trunk, the terminal parts of the common iliac veins and the start of the inferior vena cava, the obturator nerve, lumbosacral trunk and iliofemoral artery. Laterally, the inferior vena cava and the right common iliac vein lie superiorly and the right psoas major lies inferiorly. The left common iliac vein is medial to the upper part of the right common iliac artery.

Left common iliac artery The left common iliac artery is shorter than the right and is approximately 4 cm long. Lying anterior to it are the sympathetic rami to the pelvic plexus, the superior rectal artery and, at its terminal bifurcation, the ureter. The sympathetic trunk, the bodies of the fourth and fifth lumbar vertebrae and intervening disc, the obturator nerve, lumbosacral trunk and iliofemoral artery are all posterior. The left common iliac vein is posteromedial, and the left psoas major lateral, to the left common iliac artery.

Branches In addition to the external iliac and internal iliac branches, each common iliac artery also gives small branches to the peritoneum, psoas major, ureter, adjacent nerves and surrounding areolar tissue. The common iliac artery occasionally gives rise to the iliofemoral artery and accessory or replaced renal arteries if the kidney is low-lying.

The anteroposterior diameters of the common, external and internal iliac arteries are strongly correlated to body surface area (BSA) in children, in contrast to adults, in whom the relationship is inconsistent. Vessel diameters measured by ultrasonography are significantly larger in boys than in girls. The relationship between vessel diameters and BSA is shown in [Figure 73.7](#) (Munk et al 2002).

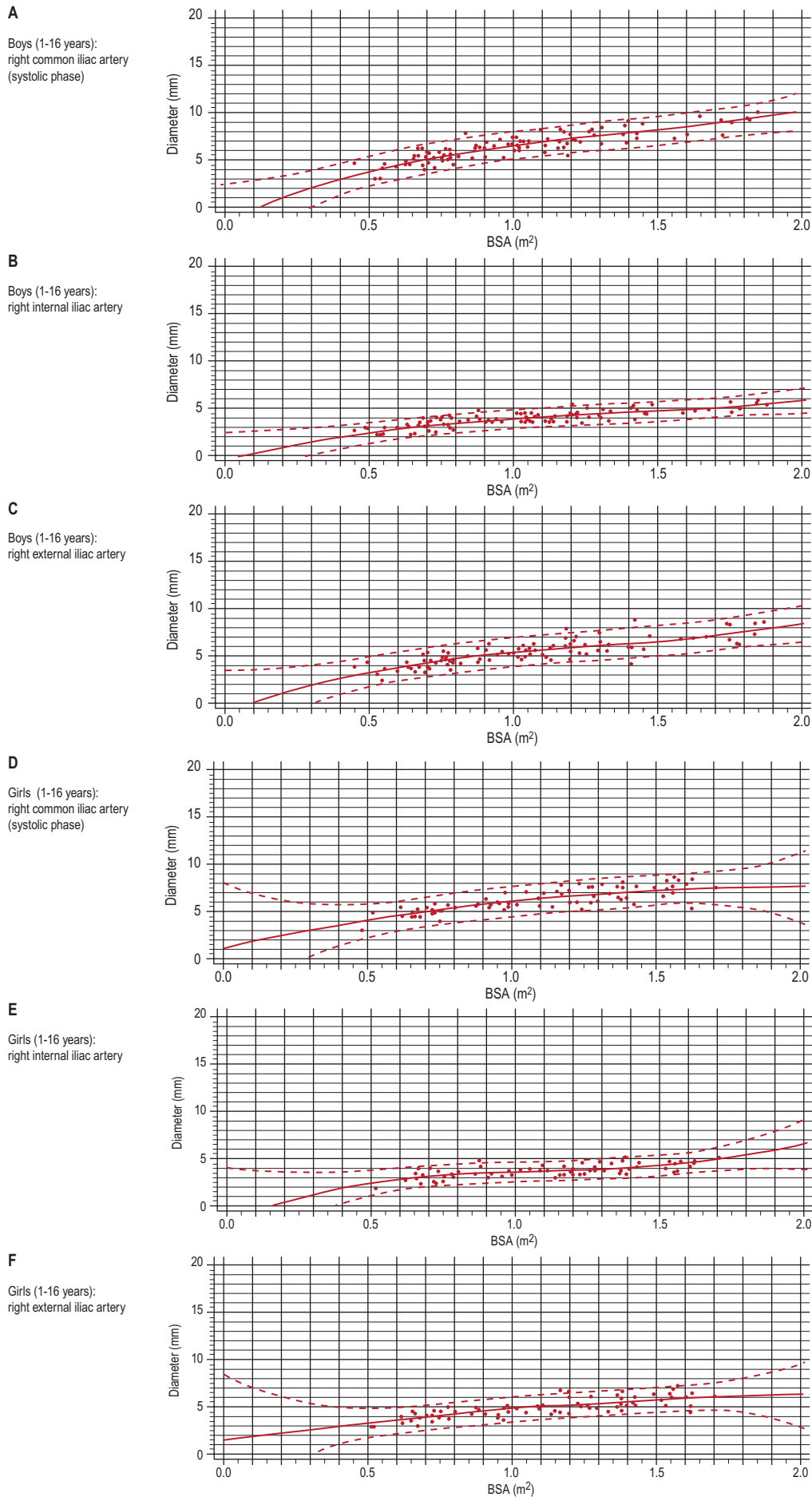


Fig. 73.7 The relationship between anteroposterior diameters of the right common, internal and external iliac arteries and body surface area (BSA) in boys and girls between the ages of 1 and 16 years. (With permission from Munk AL, Darge K, Wiesel M, Troeger J, Diameter of the infrarenal aorta and the iliac arteries in children: ultrasound measurements. Transplantation 2002;73:631–635.)

Internal iliac arteries

Each internal iliac artery, approximately 4 cm long, begins at the common iliac bifurcation, level with the lumbosacral intervertebral disc and anterior to the sacroiliac joint (Figs 73.5, 73.6). It descends posteriorly to the superior margin of the greater sciatic foramen, where it divides into an anterior trunk, which continues in the same line towards the ischial spine, and a posterior trunk, which passes back to the greater sciatic foramen. The anterior trunk primarily supplies the pelvic organs, while the posterior trunk primarily supplies muscles in the hip and back. Anterior to the artery are the ureter and, in females, the ovary and fimbriated end of the uterine tube. The internal iliac vein, lumbosacral

trunk and sacroiliac joint are posterior. Laterally are the external iliac vein, between the artery and psoas major, and the obturator nerve lying inferior to the vein. The parietal peritoneum is medial, separating it from the terminal ileum on the right and the sigmoid colon on the left, and tributaries of the internal iliac vein. For details of the considerable variation in the anatomy of the internal iliac artery, see [Roberts and Krishinger \(1967\)](#).

In the fetus, the internal iliac artery is twice the size of the external iliac artery and is the direct continuation of the common iliac artery. The main trunk ascends on the anterior abdominal wall to the umbilicus, converging on the contralateral artery, and the two arteries run

Fig. 73.5 Arteries of the male pelvis.

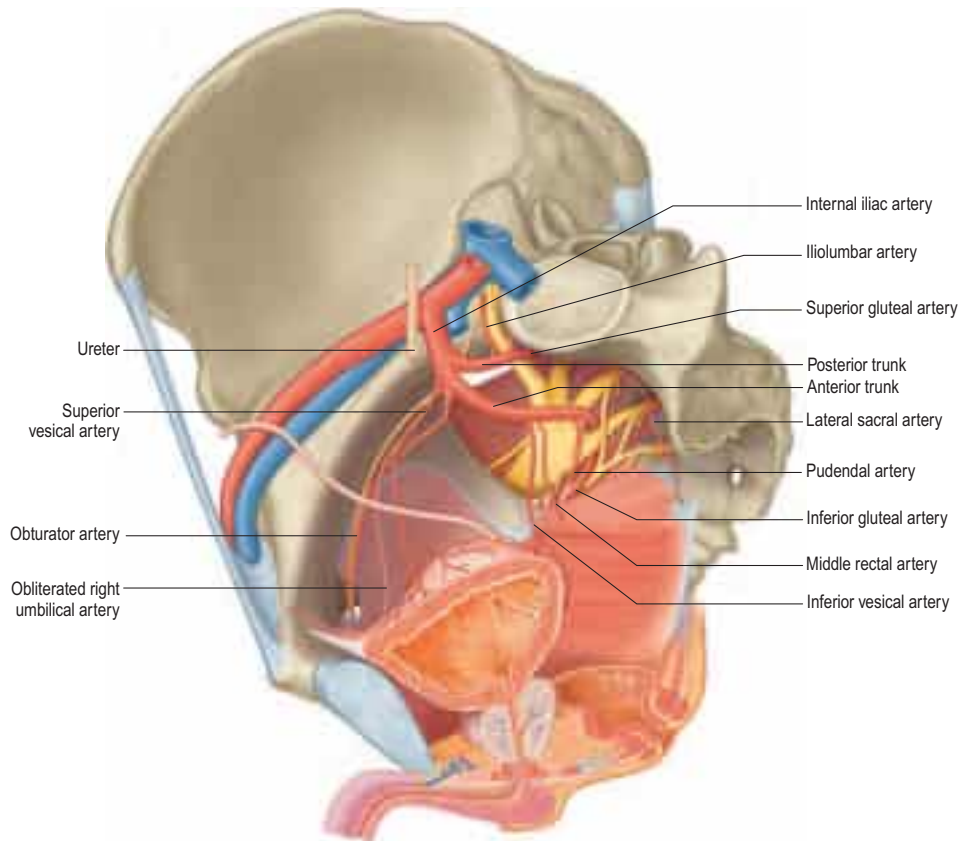
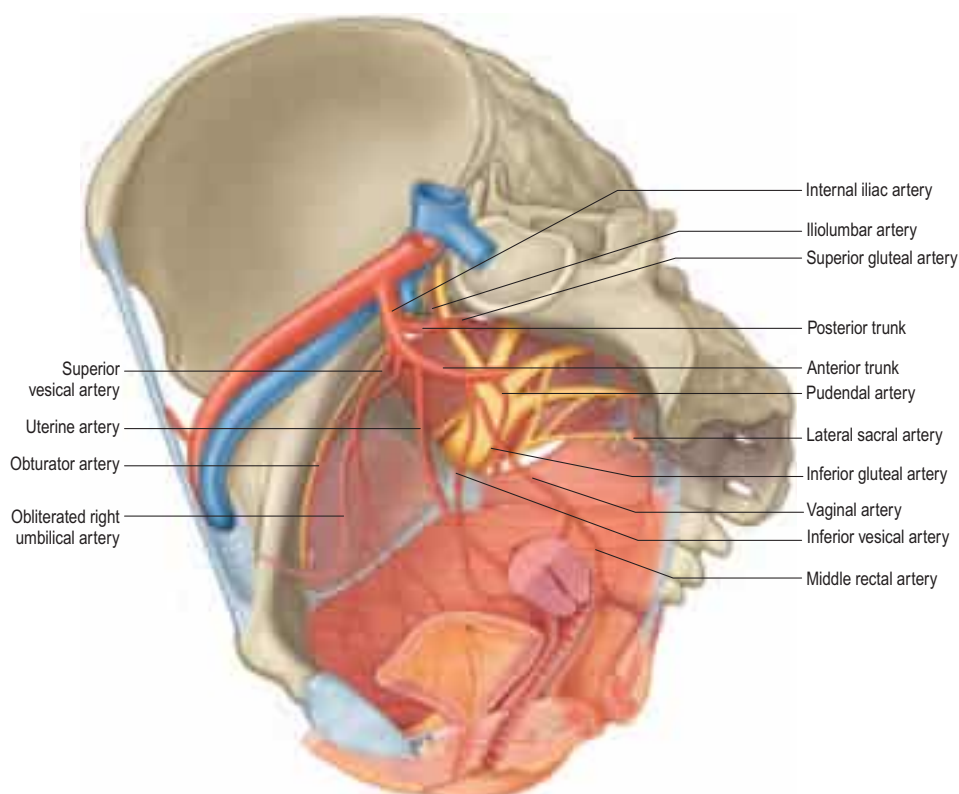


Fig. 73.6 Arteries of the female pelvis.



through the umbilicus to enter the umbilical cord as the umbilical arteries. At birth, when placental circulation ceases, only the pelvic segment remains patent as the internal iliac artery and part of the superior vesical artery; the remainder becomes a fibrous medial umbilical ligament. Persistence of the umbilical artery has been described and may cause extrinsic obstruction of the distal ureter (Gupta et al 1999). In males, the patent part (commonly, the superior vesical artery) usually gives off an artery to the vas (ductus) deferens.

Posterior trunk branches

The branches of the posterior trunk of the internal iliac artery are the iliolumbar, lateral sacral and superior gluteal arteries.

Iliolumbar artery The iliolumbar artery is the first branch of the posterior trunk and ascends laterally anterior to the sacroiliac joint and lumbosacral nerve trunk. It lies posterior to the obturator nerve and external iliac vessels, and reaches the medial border of psoas major, dividing behind it into the lumbar and iliac branches. The lumbar branch supplies psoas major and quadratus lumborum, and anastomoses with the fourth lumbar artery. It sends a small spinal branch through the intervertebral foramen between the fifth lumbar and first sacral vertebrae, to supply the cauda equina. The iliac branch supplies iliacus; between the muscle and bone, it anastomoses with the iliac branches of the obturator artery. A large nutrient branch enters an oblique canal in the ilium. Other branches run around the iliac crest, contribute to the supply of the gluteal and abdominal muscles, and anastomose with the superior gluteal, circumflex iliac and lateral circumflex femoral arteries (see Fig. 80.18).

Lateral sacral arteries The lateral sacral arteries are usually double; if they are single, they divide rapidly into superior and inferior branches. The superior and larger artery passes medially into the first or second anterior sacral foramen, supplies the sacral vertebrae and contents of the sacral canal, and then leaves the sacrum via the corresponding dorsal foramen to supply the skin and muscles dorsal to the sacrum. The inferior or lateral sacral artery crosses obliquely anterior to piriformis and the sacral anterior spinal rami, and then descends lateral to the sympathetic trunk to anastomose with its fellow and the median sacral artery anterior to the coccyx. Its branches enter the anterior sacral foramina and are distributed in the same way as the branches of the superior artery.

Superior gluteal artery The superior gluteal artery is the largest branch of the internal iliac artery and, effectively, forms the main continuation of its posterior trunk. It runs posteriorly between the lumbosacral trunk and the first sacral ramus, or between the first and second rami, and then turns slightly inferiorly, leaving the pelvis by the greater sciatic foramen above piriformis and dividing into superficial and deep branches. In the pelvis, it supplies piriformis, obturator internus and a nutrient artery to the ilium. The superficial branch enters the deep surface of gluteus maximus. Its numerous branches supply the muscle and anastomose with the inferior gluteal branches (see Fig. 80.30), while others perforate the tendinous medial attachment of the muscle to supply the skin over the sacrum, where they anastomose with the posterior branches of the lateral sacral arteries. The deep branch of the superior gluteal artery passes between gluteus medius and the bone, soon dividing into superior and inferior branches. The superior branch skirts the superior border of gluteus minimus to the anterior superior iliac spine, and anastomoses with the deep circumflex iliac artery and the ascending branch of the lateral circumflex femoral artery. The inferior branch runs through gluteus minimus obliquely, supplies it and gluteus medius, and anastomoses with the lateral circumflex femoral artery. A branch enters the trochanteric fossa to join the inferior gluteal artery and ascending branch of the medial circumflex femoral artery; other branches run through gluteus minimus to supply the hip joint.

The superior gluteal artery occasionally arises directly from the internal iliac artery with the inferior gluteal artery and sometimes from the internal pudendal artery.

Anterior trunk branches

The branches of the anterior trunk of the internal iliac artery are the superior and inferior vesical, middle rectal, vaginal, obturator, uterine, internal pudendal and inferior gluteal arteries (see Fig. 73.6). Significant variation occurs in the branching patterns of the anterior trunk; the general principles will be considered here.

Superior vesical artery The superior vesical artery is the first large branch of the anterior trunk. It lies on the lateral wall of the pelvis, just below the brim, and runs anteroinferiorly, medial to the periosteum of

the posterior surface of the pubis. It supplies the distal end of the ureter, the bladder, the proximal end of the vas deferens and the seminal vesicles. It also gives origin to the umbilical artery in the fetus, which remains as a fibrous cord – the medial umbilical ligament – in the adult. This vessel occasionally remains patent as a small artery supplying the umbilicus.

Inferior vesical artery The inferior vesical artery may arise as a common branch with the middle rectal artery. It supplies the bladder, prostate, seminal vesicles and vas deferens in the male, and the bladder in the female, where it is often replaced by the vaginal artery.

Middle rectal artery The middle rectal artery runs into the lateral fascial coverings of the mesorectum. It often consists of multiple branches, may be small, and occasionally arises either close to, or in common with, the origin of the inferior vesical artery in males.

Vaginal artery In females, the vaginal artery may replace the inferior vesical artery. It may arise from the uterine artery close to its origin and can be a single vessel or multiple branches.

Obturator artery The obturator artery runs anteroinferiorly from the anterior trunk on the lateral pelvic wall to the upper part of the obturator foramen. In the pelvis, it is related laterally to the fascia over obturator internus and is crossed on its medial aspect by the ureter and, in the male, by the vas deferens. The obturator nerve runs above the artery, the obturator vein below it. The artery provides iliac branches to the iliac fossa that supply the bone and iliacus and anastomose with the iliolumbar artery. It gives off a vesical branch that runs medially to the bladder and sometimes replaces the inferior vesical branch of the internal iliac artery. In the female, the ovary lies medial to the obturator artery. A pubic branch usually arises just before the obturator artery leaves the pelvis; it ascends over the pubis to anastomose with the contralateral artery and the pubic branch of the inferior epigastric artery.

The obturator artery leaves the pelvis via the obturator canal and divides into anterior and posterior branches that encircle the obturator foramen between obturator externus and the obturator membrane. The anterior branch curves anteriorly on the membrane and then inferiorly along its anterior margin to supply branches to obturator externus, pectineus, the femoral adductors and gracilis. It anastomoses with the posterior branch and the medial circumflex femoral artery. The posterior branch follows the posterior margin of the foramen and turns anteriorly on the ischial part to anastomose with the anterior branch. It supplies the muscles attached to the ischial tuberosity and anastomoses with the inferior gluteal artery. An acetabular branch enters the hip joint at the acetabular notch, ramifies in the fat of the acetabular fossa and sends a branch along the ligament of the femoral head.

Occasionally, the obturator artery is replaced by an enlarged pubic branch of the inferior epigastric artery that descends almost vertically to the obturator foramen. It usually lies near the external iliac vein, lateral to the femoral ring, and is rarely injured during inguinal or femoral hernia surgery. Sometimes, it curves along the edge of the lacunar part of the inguinal ligament, partly encircling the neck of a hernial sac, and may be inadvertently cut during enlargement of the femoral ring in reducing a femoral hernia.

Uterine artery The uterine artery is an additional branch in females. It is a large artery that arises below the obturator artery on the lateral wall of the pelvis and runs inferomedially into the broad ligament of the uterus (Ch. 77).

Internal pudendal artery (pelvic portion) The internal pudendal artery is the smaller terminal branch of the anterior division of the internal iliac artery. Close to its origin, it crosses anterior to piriformis, the sacral plexus and the inferior gluteal artery. It descends laterally to the inferior rim of the greater sciatic foramen, where it leaves the pelvis between piriformis and ischiococcygeus, and enters the gluteal region (see Fig. 77.3B). It next curves around the dorsum of the ischial spine and enters the ischiorectal fossa via the lesser sciatic foramen. This course effectively allows the nerve to wrap around the posterior limit of levator ani at its attachment to the ischial spine. Behind the ischial spine, the artery is covered by gluteus maximus, the pudendal nerve is medial, and the nerve to obturator internus is lateral. The artery traverses the ischiorectal fossa in Alcock's canal in the fascia covering obturator internus; it gives off an inferior rectal branch early in its course through the fossa (see Fig. 73.12). The internal pudendal artery gives off several muscular branches in the pelvis and gluteal region that supply adjacent muscles and nerves.

Inferior gluteal artery The inferior gluteal artery is the larger terminal branch of the anterior internal iliac trunk and principally supplies the buttock and thigh. It descends posteriorly, anterior to the sacral plexus and piriformis but posterior to the internal pudendal artery. It passes between either the first and second, or second and third, sacral ventral rami, then between piriformis and ischiococcygeus, before running through the lower part of the greater sciatic foramen to reach the gluteal region. The artery runs inferiorly between the greater trochanter and ischial tuberosity, together with the sciatic and posterior femoral cutaneous nerves, deep to gluteus maximus. It continues down the thigh, supplying the skin and anastomosing with branches of the perforating arteries. The inferior gluteal and internal pudendal arteries often arise as a common stem from the internal iliac artery, sometimes with the superior gluteal artery. In the pelvis, the inferior gluteal artery gives branches to piriformis, ischiococcygeus and iliococcygeus, and occasionally contributes to the middle rectal arterial supply. In the male, it may supply vessels to the seminal vesicles and prostate.

External iliac arteries

The external iliac arteries are of larger calibre than the internal iliac arteries (see Figs 73.5, 73.6). Each artery descends laterally along the medial border of psoas major, from the common iliac bifurcation to a point midway between the anterior superior iliac spine and the pubic symphysis, and enters the thigh posterior to the inguinal ligament to become the femoral artery.

The parietal peritoneum and extraperitoneal tissue separate the right external iliac artery from the terminal ileum and, usually, the appendix, and the left external iliac artery from the sigmoid colon and coils of small intestine anteromedially. At its origin, the external iliac artery may be crossed by the ureter; it is subsequently crossed by the gonadal vessels, the genital branch of the genitofemoral nerve, the deep circumflex iliac vein, and the vas deferens (male) or round ligament (female). Posteriorly, the artery is separated from the medial border of psoas major by the iliac fascia. The external iliac vein lies partly posterior to its upper part but is more medial below. Laterally, it is related to psoas major, which is covered by the iliac and psoas fasciae. Numerous lymph vessels and nodes lie on its anterior and lateral aspects.

The external iliac artery is principally the artery of the lower limb and, as such, has few branches in the pelvis. Apart from giving off very small vessels to psoas major and neighbouring lymph nodes, the artery has no branches until it gives off the deep circumflex iliac and inferior epigastric arteries, near to the point at which it passes under the inguinal ligament.

Deep circumflex iliac artery The deep circumflex iliac artery branches laterally from the external iliac artery almost opposite the origin of the inferior epigastric artery (see Fig. 78.7A). It ascends and runs laterally to the anterior superior iliac spine behind the inguinal ligament in a sheath formed by the union of the transversalis and iliac fasciae. There, it anastomoses with the ascending branch of the lateral circumflex femoral artery, pierces the transversalis fascia and skirts the internal lip of the iliac crest. About halfway along the iliac crest, it runs through transversus abdominis, and then between transversus abdominis and internal oblique, to anastomose with the iliolumbar and superior gluteal arteries. It gives off a large ascending branch at the anterior superior iliac spine that runs between internal oblique and transversus abdominis, supplies both muscles, and anastomoses with the lumbar and inferior epigastric arteries.

Inferior epigastric artery The inferior epigastric artery originates from the external iliac artery posterior to the inguinal ligament. It curves forwards in the anterior extraperitoneal tissue and ascends obliquely along the medial margin of the deep inguinal ring; from here, it continues as an artery of the anterior abdominal wall.

Veins of the pelvis

The true pelvis contains a large number of veins that drain the pelvic walls and most of the viscera contained within the pelvis, and also carry venous blood from the gluteal region, hip and thigh. The external iliac veins, which lie close to the brim of the pelvis, carry the venous drainage from most of the lower limb. There is considerable variation in the venous drainage of the pelvis: although the major veins frequently follow their named arterial counterparts, the small tributaries exhibit considerable inter-individual variation.

Common iliac veins

The common iliac vein is formed by the union of the external and internal iliac veins, anterior to the sacroiliac joints (see Fig. 78.8). It

ascends obliquely to end at the right side of the fifth lumbar vertebra, uniting at an acute angle with the contralateral vessel to form the inferior vena cava. The right common iliac vein is shorter and more nearly vertical, lying posterior, and then lateral, to its artery. The right obturator nerve passes posteriorly. The left common iliac vein is longer and more oblique, and lies first medial, then posterior, to its artery. It is crossed anteriorly by the attachment of the sigmoid mesocolon and superior rectal vessels. Each vein receives iliolumbar and, sometimes, lateral sacral veins. The left common iliac vein usually drains the median sacral vein. There are no valves in these veins.

The left common iliac vein occasionally ascends to the left of the aorta to the level of the kidney, where it receives the left renal vein and crosses anterior to the aorta to join the inferior vena cava; this vessel represents the persistent caudal half of the left postcardinal or supra-cardinal vein.

Median sacral veins The right and left medial sacral veins accompany the corresponding arteries anterior to the sacrum, where they unite to form a single vein that usually ends in the left common iliac vein, but which sometimes ends at the common iliac junction.

Internal pudendal veins The internal pudendal veins are venae comitantes of the internal pudendal artery and unite as a single vessel that drains into the internal iliac vein. They receive veins from the inferior rectal veins and either the penile bulb and scrotum (males), or the clitoris and labia (females).

Internal iliac vein

The internal iliac vein is formed by the convergence of several veins above the greater sciatic foramen. It does not have the predictable trunks and branches of the internal iliac artery but its tributaries drain the same territories. It ascends posteromedial to the internal iliac artery to join the external iliac vein, forming the common iliac vein at the pelvic brim, anterior to the lower part of the sacroiliac joint. It is covered anteromedially by parietal peritoneum. Its tributaries are the gluteal, internal pudendal and obturator veins, which originate outside the pelvis; the lateral sacral veins, which run from the anterior surface of the sacrum; and the middle rectal, vesical, uterine and vaginal veins, which originate in the venous plexuses of the pelvic viscera (Fig. 73.8).

The venous drainage of the leg may be blocked by thrombosis involving the external iliac systems and the inferior vena cava. Under these circumstances, the pelvic veins, particularly the internal iliac tributaries, enlarge and provide a major avenue of venous return from the femoral system. Surgical interference with these veins may seriously compromise venous drainage and precipitate oedema of one or both legs.

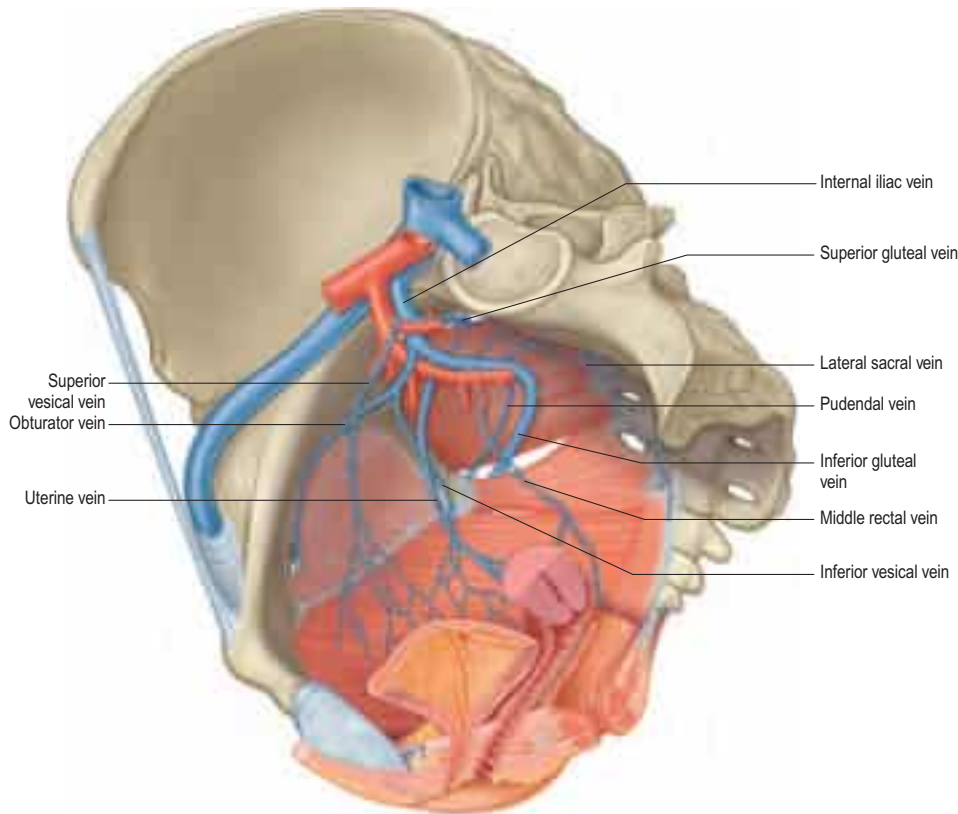
Superior gluteal veins The superior gluteal veins are the venae comitantes of the superior gluteal artery. They receive tributaries that correspond to the branches of the superior gluteal artery and enter the pelvis via the greater sciatic foramen, above piriformis. They join the internal iliac vein, frequently as a single trunk.

Inferior gluteal veins The inferior gluteal veins are venae comitantes of the inferior gluteal artery. They begin proximally and posterior in the thigh, where they anastomose with the medial circumflex femoral and first perforating veins, and they enter the pelvis low in the greater sciatic foramen, joining to form a vessel that opens into the distal (lower) part of the internal iliac vein. The inferior gluteal and superficial gluteal veins connect by perforating veins (Doyle 1970) analogous to the sural perforating veins. The gluteal veins probably have a venous 'pumping' role, and provide collaterals between the femoral and internal iliac veins.

Obturator vein The obturator vein begins in the proximal adductor region and enters the pelvis via the obturator foramen. It runs posteriorly and superiorly on the lateral pelvic wall below the obturator artery and between the ureter and internal iliac artery, and ends in the internal iliac vein. It is sometimes replaced by an enlarged pubic vein, which joins the external iliac vein.

Lateral sacral veins The lateral sacral veins accompany the lateral sacral arteries and are interconnected by a sacral venous plexus.

Middle rectal vein The middle rectal vein begins in the rectal venous plexus and drains the rectum and mesorectum. Variable in size, it runs laterally on the pelvic surface of levator ani to end in the internal iliac vein. The middle rectal vein often receives tributaries from the bladder

Fig. 73.8 Veins of the female pelvis.

and either the prostate and seminal vesicle (males) or the posterior aspect of the vagina (females).

External iliac vein

The external iliac vein is the proximal continuation of the femoral vein (see [Figs 78.8, 80.33](#)). It begins posterior to the inguinal ligament, ascends along the pelvic brim, and ends anterior to the sacroiliac joint by joining the internal iliac vein to form the common iliac vein. On the right, it lies medial to the external iliac artery, gradually inclining behind it as it ascends. On the left, it is wholly medial. Disease of the external iliac artery may cause it to adhere closely to the vein at the point where it is in contact, and, particularly on the right side, the walls of the vessels may become fused, making dissection hazardous. The external iliac vein is crossed medially by the ureter and internal iliac artery. In males, it is crossed by the vas deferens, in females by the round ligament and ovarian vessels. Psoas major lies laterally, except where the artery intervenes. The vein is usually valveless but may contain a single valve. Its tributaries are the inferior epigastric, deep circumflex iliac and pubic veins. Agenesis of the external iliac vein has been reported in association with Klippel–Trenaunay syndrome ([Dogan et al 2003](#)).

Inferior epigastric vein One or two inferior epigastric veins accompany the inferior epigastric artery and drain into the external iliac vein a little above the inguinal ligament.

Deep circumflex iliac vein The deep circumflex iliac vein is formed from venae comitantes of the corresponding artery. It joins the external iliac vein a little above the inferior epigastric vein, after crossing anterior to the external iliac artery.

Pubic vein The pubic vein connects the external iliac and obturator veins. It ascends on the pelvic surface of the pubis with the pubic branch of the inferior epigastric artery and sometimes replaces the normal obturator vein.

Lymphatic drainage of the pelvis

The lymph nodes in the pelvis are grouped around the common, external iliac and internal iliac vessels, and are named accordingly ([Nesselrod 1936](#)).

Common iliac nodes

The common iliac nodes receive the entire lymphatic drainage of the lower limb because they drain both internal and external iliac nodes.

They usually lie in medial, lateral and anterior chains around the common iliac artery, the lateral being the main route; one or two lie inferior to the aortic bifurcation and anterior to the fifth lumbar vertebra or sacral promontory. The common iliac nodes connect to the lateral aortic nodes.

External iliac nodes

The external iliac nodes usually form three subgroups, which are lateral, medial and anterior to the external iliac vessels (see [Fig. 77.3B](#)). The medial nodes are considered the main channel of drainage, collecting lymph from the lower limb via the inguinal nodes, the deeper layers of the infra-umbilical abdominal wall, the adductor region of the thigh, the glans penis or clitoris, the membranous urethra, prostate, fundus of the bladder, uterine cervix and upper vagina. Their efferents pass to the common iliac nodes (see [Fig. 78.10](#)).

Inferior epigastric and circumflex iliac nodes The inferior epigastric and circumflex iliac nodes are associated with their similarly named vessels and drain the corresponding areas to the external iliac nodes.

Internal iliac nodes

The internal iliac nodes surround the branches of the internal iliac vessels. They receive afferents from most of the pelvic viscera (with the exception of the gonads and the majority of the rectum), the deeper parts of the perineum, and the gluteal and posterior femoral muscles, and drain to the common iliac nodes. The individual groups are considered in the description of the viscera. There are frequent connections between the right and left groups, particularly when they lie close to the anterior and posterior midlines ([Fig. 73.9](#)).

INNERVATION OF THE PELVIS

The pelvis contains the lumbosacral nerve trunk, the sacral and coccygeal plexuses, and the pelvic parts of the sympathetic and parasympathetic systems. Collectively, these nerves carry the somatic and autonomic innervation to the majority of the pelvic visceral organs, and the muscles of the pelvic floor and perineum, the gluteal region and the lower limb.

The ventral rami of the sacral and coccygeal spinal nerves form the sacral and coccygeal plexuses ([Fig. 73.10](#)). The upper four sacral ventral rami enter the pelvis by the anterior sacral foramina, the fifth enters between the sacrum and coccyx, and the ventral ramus of the coccygeal nerve curves forwards below the rudimentary transverse process of the

first coccygeal segment. The first and second sacral ventral rami are large, the third to fifth diminish progressively in size, and the coccygeal ramus is the smallest. Each receives a grey ramus communicans from a corresponding sympathetic ganglion. Visceral efferent rami leave the second to fourth sacral rami as the pelvic splanchnic nerves, containing preganglionic parasympathetic fibres to minute ganglia in the walls of the pelvic viscera.

Lumbosacral trunk and sacral plexus

The sacral plexus is formed by the lumbosacral trunk, the first to third sacral ventral rami, and part of the fourth sacral ventral ramus (the remainder of the fourth sacral ventral ramus joins the coccygeal plexus).

The lumbar part of the lumbosacral trunk contains part of the fourth and all of the fifth lumbar ventral rami; it appears at the medial margin

of psoas major and descends over the pelvic brim, anterior to the sacroiliac joint, to join the first sacral ramus. The greater part of the second and third sacral rami converge on the inferomedial aspect of the lumbosacral trunk in the greater sciatic foramen to form the sciatic nerve. The ventral and dorsal divisions of the nerves do not separate physically from each other but their fibres remain separate within the rami; the ventral and dorsal divisions of each contributing root join within the sciatic nerve. The fibres of the dorsal divisions will go on to form the common fibular nerve, and the fibres of the ventral division form the tibial nerve. The sciatic nerve, occasionally, divides into common fibular and tibial nerves inside the pelvis; when this occurs, the common fibular nerve usually runs through piriformis.

The sacral plexus lies against the posterior pelvic wall anterior to piriformis, posterior to the internal iliac vessels and ureter, and behind the sigmoid colon on the left. The superior gluteal vessels run either between the lumbosacral trunk and first sacral ventral ramus, or between the first and second sacral rami, while the inferior gluteal vessels lie between either the first and second, or second and third, sacral rami.

The sacral plexus is not commonly involved in malignant tumours of the pelvis because it lies behind the relatively dense presacral fascia, which resists all but locally very advanced malignant infiltration. When it occurs, there is intractable pain in the distribution of the branches of the plexus, which may be very difficult to treat. The plexus may also be involved in the reticuloses or be affected by plexiform neuromas.

Branches of the sacral plexus

The branches of the sacral plexus are shown in Table 73.1.

The course and distribution of most of the branches of the sacral plexus are covered fully in Section 9.

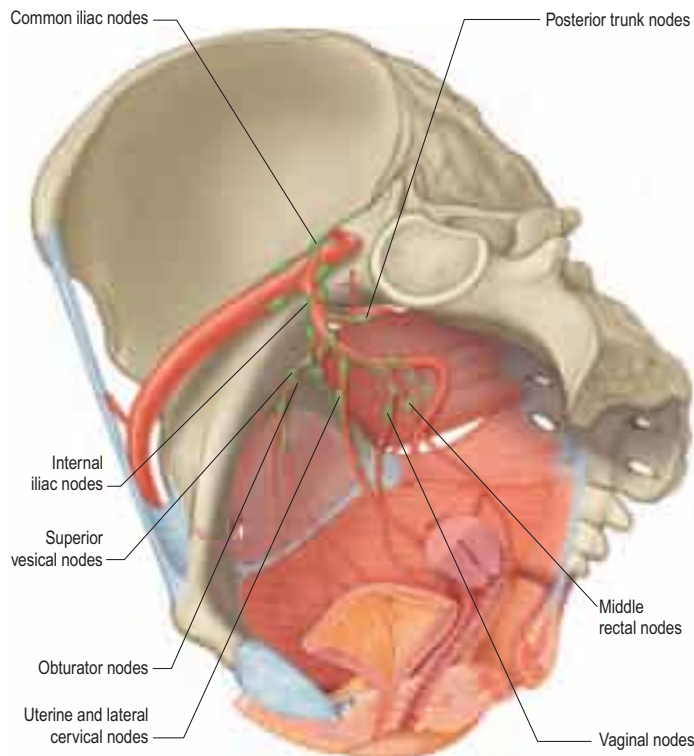
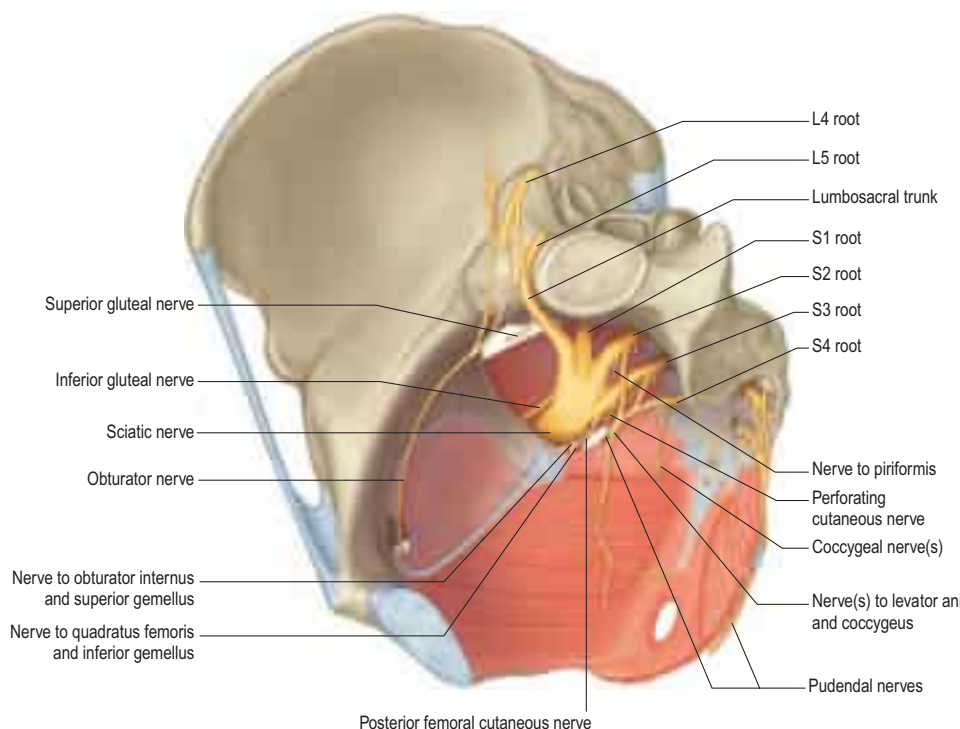


Fig. 73.9 Lymphatic drainage of the female pelvis and urinary bladder.

Table 73.1 Branches of the sacral plexus

Nerve	Ventral divisions	Dorsal divisions
Nerve to quadratus femoris and gemellus inferior	L4, 5; S1	
Nerve to obturator internus and gemellus superior	L5; S1, 2	
Nerve to piriformis		S2 (S1)
Superior gluteal nerve		L4, 5; S1
Inferior gluteal nerve		L5; S1, 2
Posterior femoral cutaneous nerve	S2, 3	S1, 2
Tibial (sciatic) nerve	L4, 5; S1, 2, 3	
Common fibular (sciatic) nerve		L4, 5; S1, 2
Perforating cutaneous nerve		S2, 3
Pudendal nerve	S2, 3, 4	
Nerves to levator ani and external anal sphincter	S4	

Fig. 73.10 The lumbosacral plexus in the pelvis.



Pudendal nerve (in the pelvis)

The pudendal nerve arises from the ventral divisions of the second, third and fourth sacral ventral rami and is formed just above the superior border of the sacrotuberous ligament and the upper fibres of ischiococcygeus (Klink 1953, Sato 1980). It leaves the pelvis via the greater sciatic foramen between piriformis and ischiococcygeus (see Fig. 73.10), enters the gluteal region and passes dorsal to the sacrospinous ligament, close to its attachment to the ischial spine, where it lies medial to the internal pudendal vessels. It accompanies the internal pudendal artery through the lesser sciatic foramen into the pudendal (Alcock's) canal on the lateral wall of the ischio-anal fossa. In the posterior part of the canal, it gives rise to the inferior rectal and perineal nerves; the dorsal nerve of the penis or clitoris continues ventrally from this origin.

Sacral visceral branches

Visceral branches – the pelvic splanchnic nerves – arise from the second to fourth sacral ventral rami and innervate the pelvic viscera.

Sacral muscular branches

Several muscular branches arise from the fourth sacral ventral ramus to supply the superior surface of levator ani and the upper part of the external anal sphincter. The branches to levator ani enter the superior (pelvic) surface of the muscle whilst the branch to the external anal sphincter (also referred to as the perineal branch of the fourth sacral nerve) reaches the ischio-anal fossa by running either through ischiococcygeus, or between ischiococcygeus and iliococcygeus. It supplies the skin between the anus and coccyx via its cutaneous branches.

Coccygeal plexus

The coccygeal plexus is formed by a small descending branch from the fourth sacral ramus and by the fifth sacral and coccygeal ventral rami. The fifth sacral ventral ramus emerges from the sacral hiatus, curves round the lateral margin of the sacrum below its cornu, and pierces ischiococcygeus from below to reach its upper, pelvic, surface. Here it is joined by a descending branch of the fourth sacral ventral ramus; the small trunk so formed descends on the pelvic surface of ischiococcy-

geus, joins the minute coccygeal ventral ramus that emerges from the sacral hiatus, and curves round the lateral coccygeal margin, piercing coccygeus to reach the pelvis. The small trunk that is formed in this way is the coccygeal plexus. Anococcygeal nerves arise from it and form a few fine filaments that pierce the sacrotuberous ligament to supply the adjacent skin.

Pelvic part of the sympathetic system

The pelvic sympathetic trunk lies in the extraperitoneal tissue, anterior to the sacrum beneath the presacral fascia (Fig. 73.11), and supplies sympathetic innervation to the pelvic organs and vascular system. It lies medial or anterior to the anterior sacral foramina and has four or five interconnected ganglia. Above, it is continuous with the lumbar sympathetic trunk. The right and left trunks converge below the lowest ganglia and unite in the small ganglion impar anterior to the coccyx. Grey rami communicantes pass from the ganglia to sacral and coccygeal spinal nerves but there are no white rami communicantes. Medial branches connect across the midline, and twigs from the first two ganglia join the inferior hypogastric plexus or the hypogastric 'nerve'. Other branches form a plexus on the median sacral artery.

Vascular branches

Preganglionic fibres for the vessels supplying the pelvis and lower limb are derived from the lower three thoracic and upper two or three lumbar spinal segments. They reach the lower thoracic and upper lumbar ganglia through white rami communicantes and descend through the sympathetic trunk to synapse in the lumbar ganglia. Postganglionic fibres pass from these ganglia via grey rami communicantes to the femoral nerve, which carries them to the femoral artery and its branches. Some fibres descend through the lumbar ganglia to synapse in the upper two or three sacral ganglia, from which postganglionic axons pass through grey rami communicantes to the roots of the sacral plexus. Those in the pudendal and superior and inferior gluteal nerves accompany the arteries of the same name to the gluteal and perineal tissues; branches may also supply the pelvic lymph nodes. Those joining the tibial nerve are carried to the popliteal artery and distributed via its branches to the leg and foot.

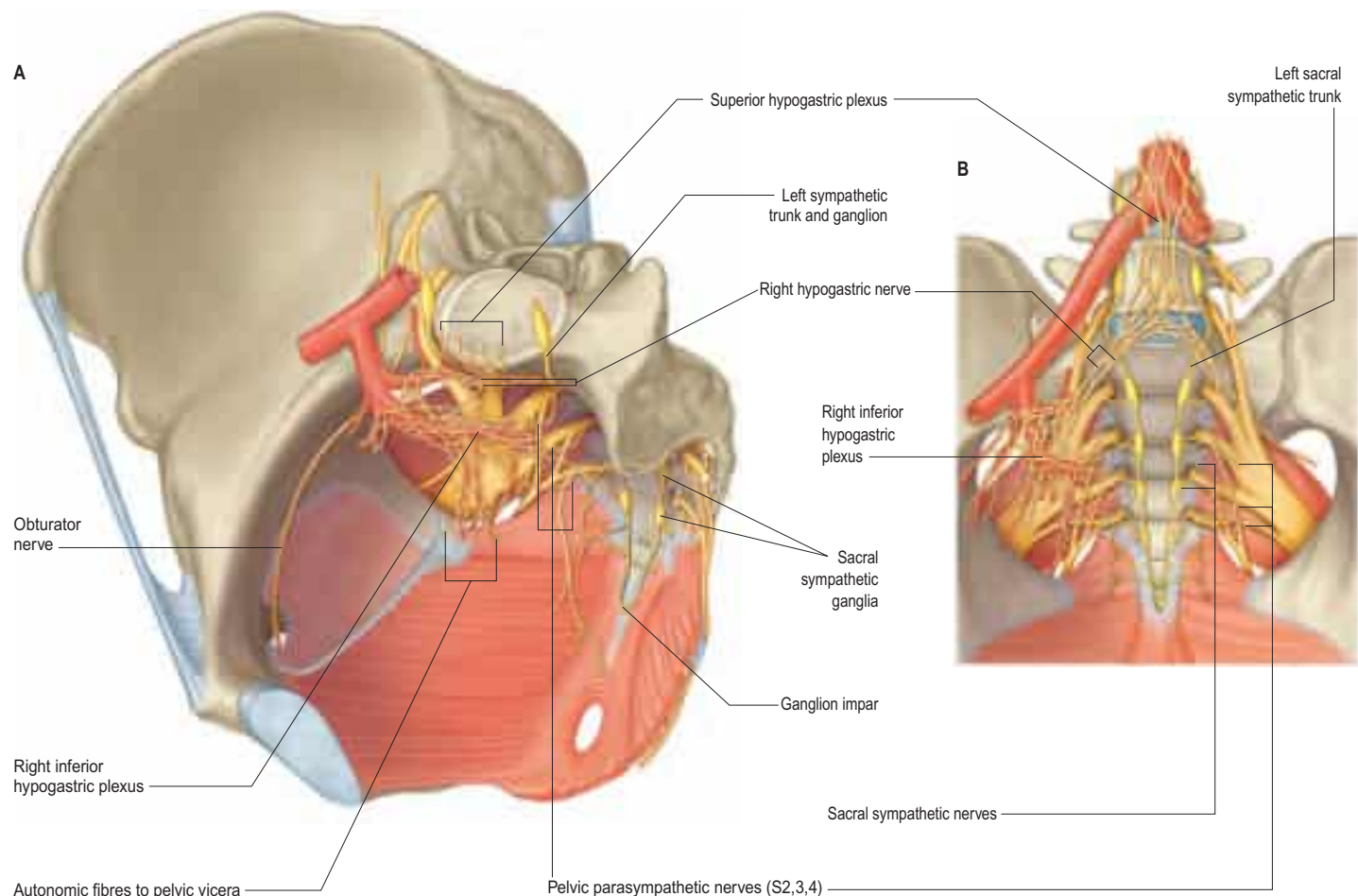


Fig. 73.11 Autonomic nerves of the pelvis.

Sympathetic denervation of vessels in the lower limb can be effected by removing or ablating the upper three lumbar ganglia and the intervening parts of the sympathetic trunk, which is used, rarely, in treating vascular insufficiency of the lower limb.

PERINEUM

MUSCLES AND FASCIAE OF THE PERINEUM

The perineum is an approximately diamond-shaped region that lies below levator ani, between the inner aspects of the thighs and anterior to the sacrum and coccyx. It is usually described as if from the position of an individual lying supine with the hip joints in abduction and partial flexion. The surface projection of the perineum and the form of the skin covering it vary considerably, depending on the position of the thighs, whereas the deep tissues themselves occupy relatively fixed positions. The perineum is bounded anteriorly by the pubic symphysis and its arcuate ligament, posteriorly by the coccyx, anterolaterally by the ischiopubic rami and the ischial tuberosities, and posterolaterally by the sacrotuberous ligaments. The deep limit of the perineum is the inferior surface of the pelvic diaphragm, and its superficial limit is the skin that is continuous with that over the medial aspect of the thighs and the lower abdominal wall. An arbitrary line joining the ischial tuberosities (the inter-ischial line) divides the perineum into an anterior urogenital triangle and a posterior anal triangle. The urogenital triangle faces downwards and forwards, whereas the anal triangle faces downwards and backwards at an approximate angle of 120° from the plane of the urogenital triangle.

The male urogenital triangle contains the bulb and attachments of the penis (Fig. 73.12) (Ch. 76), and the female urogenital triangle contains the mons pubis, the labia majora, the labia minora, the clitoris and the vaginal and urethral orifices (Ch. 77).

Anal triangle

The structure of the anal triangle is similar in males and females, the main difference reflecting the wider transverse dimension of the triangle in females that is associated with giving birth. The anal triangle contains the anal canal and its sphincters, and the ischio-anal fossa and its contained nerves and vessels. It is lined by superficial and deep fascia.

Superficial fascia of the anal triangle

The superficial fascia (subcutaneous tissue; tela subcutanea) of the region is thin and is continuous with the superficial/subcutaneous fascia of the skin of the perineum, thighs and buttocks.

Deep fascia of the anal triangle

The deep fascia (fascia musculorum) lines the inferior surface of levator ani and is continuous at its lateral origin with the fascia over obturator internus below the attachment of levator ani. It lines the deep portion of the ischio-anal fossa and its lateral walls.

Ischio-anal fossa

The ischio-anal fossa is an approximately horseshoe-shaped region filling the majority of the anal triangle. Although often referred to as a space, it is filled with loose adipose tissue and occasional blood vessels and nerves (see Fig. 73.12). The 'arms' of the horseshoe are triangular in cross-section because levator ani slopes medially from its lateral pelvic origin towards the anorectal junction (see Fig. 66.44). The anal canal and its sphincters lie in the centre of the horseshoe. Above them, the medial limit of the fossa is formed by the deep fascia over levator ani. The outer boundary of the fossa is formed anterolaterally by the fascia over obturator internus and the periosteum of the ischial tuberosities. Posterolaterally, the outer boundary is formed by the lower border of gluteus maximus and the sacrotuberous ligament.

There is an anterior recess to the ischio-anal fossa that lies cranial to the perineal membrane and transverse perineal muscles. It extends anteriorly as far as the posterior surface of the pubis, below the attachment of levator ani. Posteriorly, the fossa contains the attachment of the external anal sphincter to the tip of the coccyx; above and below this, the adipose tissue of the fossa is uninterrupted across the midline. These continuations of the ischio-anal fossa mean that infections, tumours and fluid collections within not only may enlarge relatively freely to the side of the anal canal, but also may spread with little resistance to the contralateral side and deep to the perineal membrane. The internal pudendal vessels and accompanying nerves lie in the lateral wall of the ischio-anal fossa, enclosed in fascia forming the pudendal canal. The inferior rectal vessels and nerves cross the fossa from the pudendal canal and often branch within it.

The ischio-anal fossa is an important surgical plane during resections of the anal canal and anorectal junction for malignancy. It provides an easy, relatively bloodless, plane of dissection that encompasses all of the muscular structures of the anal canal and leads to the inferior surface of levator ani, through which the dissection is carried.

External anal sphincter

The external anal sphincter is a band of striated muscle that surrounds the lowest part of the anal canal (Oh and Kark 1972, Dalley 1987, Lawson 1974b). The uppermost (deepest) fibres blend with the lowest fibres of puborectalis; the two are seen to be contiguous on endoanal ultrasound and magnetic resonance imaging. Anteriorly, some of these upper fibres decussate into the superficial transverse perineal muscles. Posteriorly, fibres are attached to the anococcygeal raphe. The majority

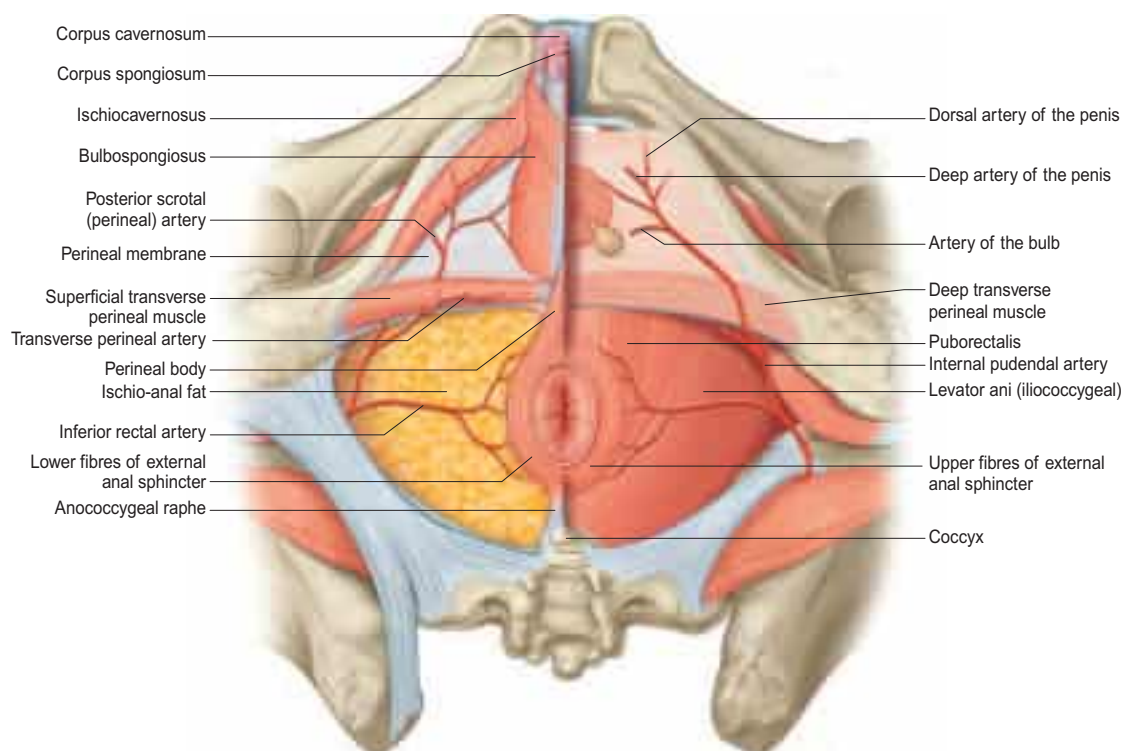


Fig. 73.12 Muscles and fasciae of the male perineum. On the left side, the skin and superficial fascia of the perineum only have been removed. The posterior scrotal (perineal) artery has been shown as it runs forwards into the scrotal tissues. On the right side, the corpora cavernosa and corpus spongiosum and their associated muscles, the superficial perineal muscles and perineal membrane have been removed to reveal the underlying deep muscles and arteries of the perineum. All veins and nerves have been omitted for clarity.

of the middle fibres of the external anal sphincter surround the lower part of the internal sphincter and are attached anteriorly in the perineal body, and posteriorly to the coccyx via the anococcygeal ligament. Some fibres from each side of the sphincter decussate in these areas to form a sort of commissure in the anterior and posterior midline. The anterior and posterior attachments of the external anal sphincter give the muscular tube an oval profile lying anteroposteriorly. A subcutaneous portion encircles the anal verge and creates the radial skin creases surrounding the anus. The lower fibres lie below the level of the internal anal sphincter and are separated from the lowest anal epithelium by submucosa. The thickness of the external anal sphincter in children is positively correlated with age (de la Portilla and López-Alonso 2009, Rehman et al 2011).

Anococcygeal ligament and iliococcygeal raphe

The anococcygeal ligament is a musculotendinous structure running between the middle portion of the external anal sphincter and the coccyx. The iliococcygeal raphe (the decussation of the posterior fibres of iliococcygeus) lies just above the anococcygeal ligament and is separated from the rectum by presacral fascia. These structures are often referred to as the postanal plate. Division of the anococcygeal raphe may cause descent of the anal canal and a lowering of the posterior part of the anal triangle, but does not demonstrably interfere with the process of defecation.

Urogenital triangle

The urogenital triangle is bounded posteriorly by the inter-ischial line, which usually overlies the posterior border of the transverse perineal muscles. Anteriorly and laterally, it is bounded deeply by the pubic symphysis and ischiopubic rami. In males, the urogenital triangle extends superficially to encompass the scrotum and the root of the penis. In females, it extends to the lower limit of the labia and mons pubis. The urogenital triangle is divided into two parts by a strong perineal membrane: the deep perineal space lies above the membrane, and the superficial perineal space lies below it.

The female urogenital triangle includes muscles, fasciae, erectile structures and spaces similar to those in the male. There are some differences in size and disposition caused by the presence of the vagina and female external genitalia.

Deep perineal space

Perineal membrane

For many years, it was thought that the anterior pelvic outlet was spanned by a triangular, trilaminar, musculofascial structure – the uro-

genital diaphragm – consisting of urogenital sphincter muscles sandwiched between two layers of fascia and connected to the perineal body, the periosteum of the ischiopubic rami and the arcuate ligament of the pubis, and perforated by the urethra and vagina. Oelrich introduced the alternative term ‘perineal membrane’ (Oelrich 1983). Recent detailed histological examination of serial cross-sections supports the concept of the perineal membrane as part of a larger interconnected support apparatus. It has distinct dorsal and ventral portions that are intimately connected with levator ani: the dorsal portion is related to the support of the perineal body and lateral vaginal wall by its attachment to the ischiopubic ramus, and the ventral portion is contiguous with the urethral supportive apparatus (Stein and DeLancey 2008). The perineal membrane is particularly thick where it is attached to the arcuate ligament of the pubis, and is here referred to as the transverse perineal ligament. The posterior border of the perineal membrane is continuous with the deep part of the perineal body at its dorsal margin, and is continuous with the fascia over the superficial transverse perineal muscles.

In the male, the perineal membrane is crossed by several structures: the urethra, which traverses it 2–3 cm behind the inferior border of the pubic symphysis; the vessels and nerves to the bulb of the penis; the ducts of the bulbourethral glands, posterolateral to the urethral orifice; the deep dorsal vessels and dorsal nerves of the penis, behind the pubic arch in the midline; and the posterior scrotal vessels and nerves, anterior to the transverse perineal.

In the female, the perineal membrane is divided almost into two halves by the vagina and urethra, such that it forms a triangle on each side of these structures. The lateral margins of the vagina are attached to the perineal membrane at the level of the hymenal ring, and levator ani lies on its cranial surface. The ducts of Bartholin’s glands are at this level in the posterior lateral introitus. The deep dorsal vessels and dorsal nerves of the clitoris lie within its fibres.

Urethral sphincter mechanism

The urethral sphincter mechanism consists of both striated and smooth muscle sphincters (Oelrich 1980, Oelrich 1983, Huisman 1983). The striated urogenital sphincter has an upper circular element that surrounds the urethra in the female and the apex of the prostate in the male, between the vesical neck and perineal membrane. In the female, at the level of the perineal membrane, it extends laterally outside of the urethra in two arch-shaped bands that lie on the cranial surface of the perineal membrane (Fig. 73.13). One, the compressor urethrae, follows the pubic arch to attached connective tissue in this area near the inner surface of the ischiopubic ramus. The other, the urethrovaginal sphincter, extends

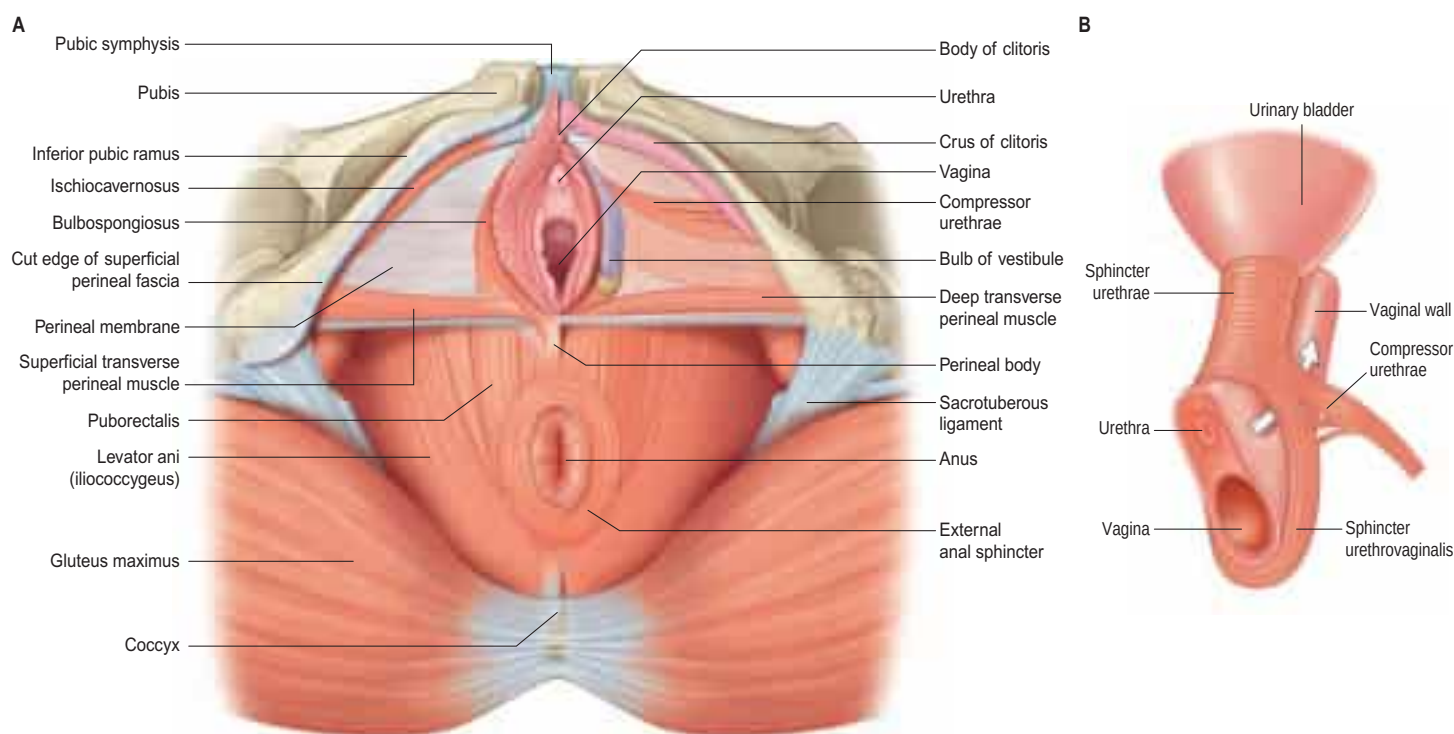


Fig. 73.13 Muscles of the female perineum. **A**, On the right side, the membranous layer of superficial fascia has been removed (note the cut edge). On the left side, superficial perineal muscles and overlying fascia have been removed to show the deep perineal muscles. **B**, The continuity of the deep perineal muscles with sphincter urethrae.

caudally to surround the lateral margin of the vaginal wall. In the male, these lateral extensions are not well developed and the sphincter in this area primarily encircles the membranous urethra, forming a robust constrictor at this level. The smooth muscle sphincter consists of circularly orientated smooth muscle cells that lie between the striated sphincter and the longitudinal smooth muscle and urethral lumen. It is not well developed. In children, the inner diameter of the urethral sphincter increases with age: a cutoff value of 3 mm has been reported to predict detrusor–sphincter incoordination (Kakizaki et al 2003).

Innervation The urethral sphincter mechanism extends from the perineum through the urogenital hiatus into the pelvic cavity. It probably receives innervation via the perineal branch of the pudendal nerve from below, and direct branches from the sacral plexus and the pelvic splanchnic nerves from above. All these nerves originate in the second, third and fourth sacral spinal segments.

Actions Urinary incontinence is primarily a problem that affects women and is more common after childbirth. Continence in women is maintained by an elegantly orchestrated system of muscles and connective tissue that involves the urethral sphincters and the tissues that support them. The urethral sphincter mechanism compresses the mid-urethra in females and membranous urethra in males, particularly when the bladder contains fluid. Its location around the region of highest urethral closing pressure suggests that it plays an important role in the continence of urine. At rest, activity of both the smooth and striated sphincters contributes to this closure and, during times of increased need, voluntary contraction of the striated sphincter augments this closure. In the distal portion of the urethra, where it is adjacent to the perineal membrane, arch-shaped components of the striated sphincter (compressor urethrae and urethrovaginal sphincter) pass between the urethra and pubic bone, and contraction compresses the lumen until it is closed. All muscles are relaxed during micturition to aid voiding, but striated muscle contraction can help to expel final drops of urine, or of semen in the male, from the bulbar urethra.

During increases in abdominal pressure that occur, for example, during a cough, bladder pressure rises above the normal urethral closure pressure that keeps the lumen of the urethra closed. In this situation, the rise in urethral closure pressures (called pressure transmission) that prevents urine leakage occurs because the urethra is compressed against the fascial tissues that lie between it and the vaginal wall. If this layer is stable and unyielding, the urethra can be compressed closed against it, but if it is not well supported, this mechanism is less effective and leakage can occur (DeLancey 1994). Overall continence during such events is a combination of sphincter constriction due to the actions of the muscles in the urethral wall and pressure transmission; sphincter function is the predominant factor (DeLancey et al 2008).

Superficial and deep perineal fasciae

Superficial perineal fascia

The tissue commonly referred to as the superficial fascia of the perineum (Colles' fascia) forms a clear, surgically recognizable, plane beneath the skin of the anterior perineum (Tobin and Benjamin 1949). It is firmly attached posteriorly to the fascia over the superficial transverse perineal muscles and the posterior limit of the perineal membrane. Laterally, it is attached to the margins of the ischiopubic rami as far back as the ischial tuberosities. From here, it runs more superficially to the skin of the urogenital triangle, lining the skin of the external genitalia; in the male, it is also continuous with the fascial layer in the skin of the scrotum that contains the dartos muscle. In females, the fascia follows the same limits but is much less extensive in the labia majora. This layer runs anteriorly and superiorly into the skin of the lower abdominal wall where it is continuous with the membranous fascia (Scarpa's fascia). English-speaking anatomists refer to these tissue layers as a 'superficial fascia' but this is a misnomer; more accurately, they constitute the membranous layer of subcutaneous tissue (tela subcutanea).

Deep perineal fascia (investing fascia of the superficial perineal muscles)

The tissue commonly referred to as the deep perineal fascia is a layer of fascia that overlies the superficial muscles of the perineum (bulbospongiosus, ischiocavernosus, superficial transverse perineal muscles). It is, in effect, the investing fascia of the superficial perineal muscles and, as such, is firmly attached to the borders of the muscles at attachments to the ischiopubic rami, posterior margin of the perineal membrane and perineal body. Anteriorly, it fuses with the suspensory

ligament of the penis or clitoris, and the fasciae of external oblique and the rectus sheath.

Superficial and subcutaneous perineal pouches

Subcutaneous perineal pouch

The subcutaneous perineal pouch lies between the deep perineal fascia and the superficial perineal fascia. Under normal circumstances, these two layers are only separated by relatively thin subcutaneous connective tissue; the skin of the anterior perineum and external genitalia is relatively mobile over the deeper structures. However, this pouch is capable of expanding considerably in the presence of fluid accumulation; blood, urine or fluid collecting in the subcutaneous pouch following trauma or surgery on the urogenital triangle will spread throughout the tissues of the triangle, including the scrotum or labia majora, but cannot pass posteriorly into the anal triangle or laterally into the medial thigh because of the firm tethering of the posterior attachments of the subcutaneous fascia. Since the superficial perineal fascia is in continuity with the fascia of the anterior abdominal wall, fluid, blood or pus may also track freely between the subcutaneous tissues of the anterior abdominal wall and the subcutaneous perineal pouch (e.g. postsurgical haematomas of the abdominal wall readily cause discolouration of the perineal and genital skin).

Superficial perineal pouch

The superficial perineal pouch lies below the perineal membrane and is limited superficially by the deep perineal fascia (investing fascia of the superficial perineal muscles) (Fig. 73.14). It contains the corpora cavernosa and corpus spongiosum, ischiocavernosus, bulbospongiosus and the superficial transverse perineal muscles, and branches of the pudendal vessels and nerves. In the female, it is crossed by the urethra and vagina and contains the clitoris. In the male, it contains the urethra as it runs in the root of the penis. It is a fully confined space; injuries to the contents of the space (such as bleeding into the urethra in the penile root) do not communicate with the deep or subcutaneous pouches unless the fascial coverings are also lacerated or breached.

Perineal body

The perineal body is not a structure but an aggregation of fibromuscular tissue located in the midline at the junction between the anal and urogenital triangles, just ventral to the anal sphincter (Oh and Kark 1973). It is attached to many structures in both the deep and superficial urogenital spaces. Posteriorly, it merges with fibres from the middle part of the external anal sphincter and the conjoint longitudinal coat. Superiorly, it is continuous with the rectoprostatic or rectovaginal septum, including fibres from levator ani (puborectalis or pubovaginalis). Anteriorly, it receives a contribution from the deep and superficial transverse perineal muscles and bulbospongiosus (see Figs 73.4, 73.13). The perineal body is continuous with the perineal membrane and the superficial perineal fascia. Since the latter runs forwards into the skin of the perineum, the perineal body is tethered to the central perineal skin, which is often puckered over it. In males, this is continuous with the perineal raphe in the skin of the scrotum. In females, the perineal body lies directly posterior, and is attached, to the posterior commissure of the labia majora and the introitus of the vagina.

Spontaneous lacerations of the perineal body sustained during childbirth are often associated with damage to the anterior fibres of the external anal sphincter. The deliberate division of the perineal body to facilitate delivery (episiotomy) is sometimes angled laterally to avoid such sphincteric injuries. The perineal body is often used to position radiological markers in the assessment of pelvic floor dysfunction.

Superficial transverse perineal muscles

The superficial transverse perineal muscles are narrow strips of muscle that run more or less transversely across the superficial perineal space anterior to the anus, from the medial and anterior aspects of the ischial tuberosities to the perineal body (see Fig. 73.13). A few fibres may also pass into the ipsilateral bulbospongiosus or external anal sphincter. They are occasionally small and may be absent.

Bulbospongiosus

Bulbospongiosus differs between the sexes. In the male, it lies in the midline, anterior to the perineal body (see Fig. 73.12). It consists of two symmetrical parts united by a median fibrous raphe. The fibres attach to the perineal body, in which they decussate, and to the transverse superficial perineal and the external anal sphincter; they diverge like the sides of a feather from the median raphe. A thin layer of posterior fibres joins the posterior portion of the perineal membrane. The

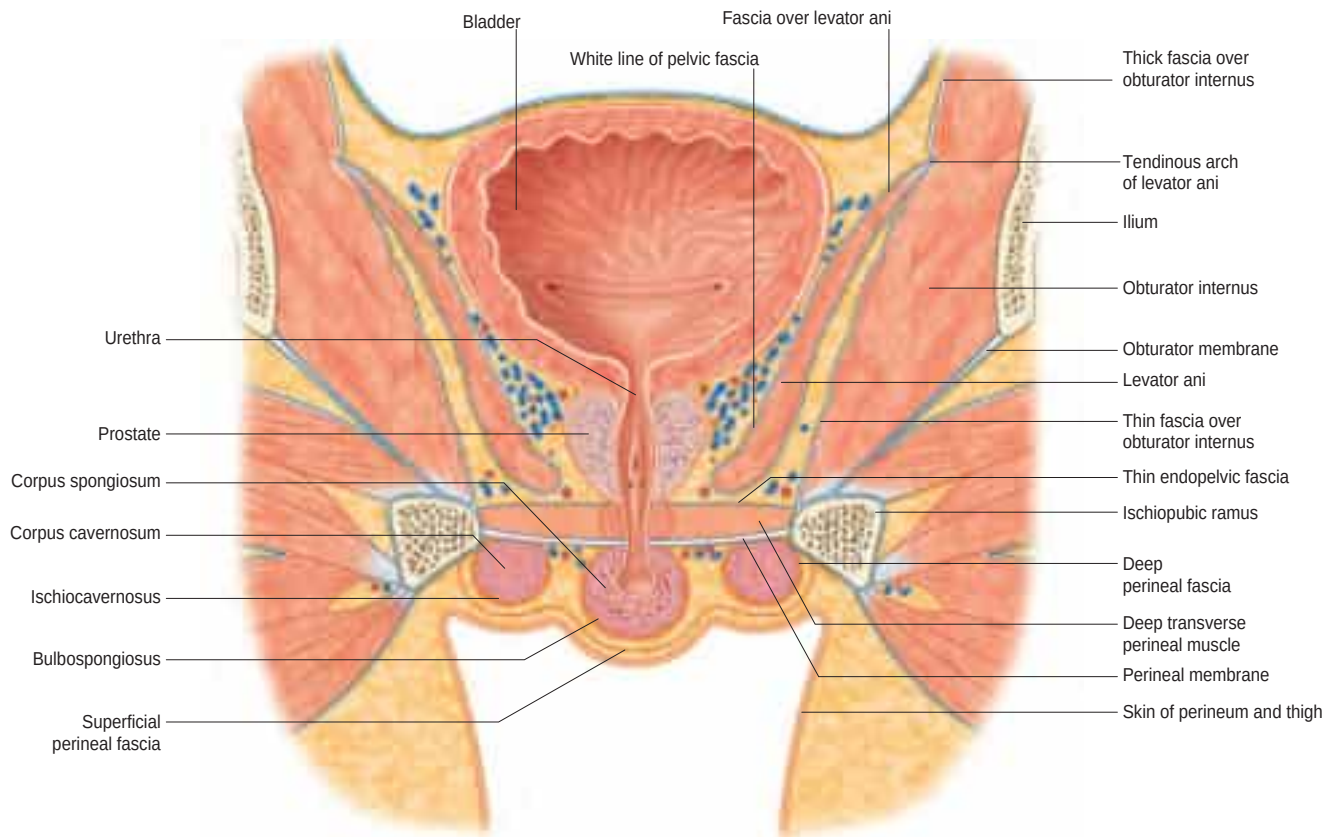


Fig. 73.14 Muscles and fasciae of the male perineum: coronal view. The section passes through the bulb of the penis at the level of the urethra. The deep perineal space is continuous with the ischio-anal fossa posteriorly. The visceral and parietal fasciae have been omitted for clarity. The pelvic fascia over the 'pelvic' aspect of the deep transverse perineal muscle is very thin and does not form a distinct layer: in places it blends with the parietal pelvic fascia over the inferior aspect of levator ani.

majority of the middle fibres encircle the bulb of the penis and adjacent corpus spongiosum, and attach to an aponeurosis on the dorsal surfaces. The anterior fibres spread out over the sides of the corpora cavernosa, ending partly in them, anterior to ischiocavernosus, and partly in a tendinous expansion that covers the dorsal vessels of the penis. In the female, bulbospongiosus also attaches to the perineal body, but the muscle on each side is separate and covers the superficial parts of the vestibular bulbs and greater vestibular glands (see Fig. 73.13). Fibres run anteriorly on either side of the vagina to attach to the corpora cavernosa clitoridis, and a few fibres cross over the dorsum of the body of the clitoris.

Actions In the male, bulbospongiosus helps to empty the urethra of urine after the bladder has emptied. It may assist in the final stage of erection as the middle fibres compress the erectile tissue of the bulb and the anterior fibres contribute by compressing the deep dorsal vein of the penis. It contracts six or seven times during ejaculation, assisting in the expulsion of semen. In the female, bulbospongiosus acts to constrict the vaginal orifice and express the secretions of the greater vestibular glands. Anterior fibres contribute to erection of the clitoris by compressing its deep dorsal vein.

Ischiocavernosus

In the male, ischiocavernosus covers the crus penis. It is attached by tendinous and muscular slips to the medial aspect of the ischial tuberosity posteriorly, and to the ischial ramus on both sides of the crus (see Fig. 73.12). These fibres end in an aponeurosis that is attached to the sides and undersurface of the crus penis. In the female, ischiocavernosus is related to the crus of the clitoris but is otherwise similar to the corresponding muscle in the male (Figs 73.15–73.16).

Actions Ischiocavernosus compresses the crus penis in males and may help to maintain penile erection. The muscles form a triangle on each side of the midline with bulbospongiosus medially and the superficial transverse perineal muscles posteriorly, attached to the perineal membrane; when contracted, the two ischiocavernosi act together to stabilize the erect penis. In the female, ischiocavernosus may help to promote increased pressure in the clitoris.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE OF THE PERINEUM

Arteries of the perineum

Internal pudendal artery (in the perineum)

The internal pudendal artery enters the perineum around the posterior aspect of the ischial spine and runs on the lateral wall of the ischio-anal fossa in the pudendal (Alcock's) canal with the pudendal veins and the pudendal nerve. The canal lies about 4 cm above the lower limit of the ischial tuberosity and is formed by connective tissue binding the vessels and nerve to the medial surface of the fascia covering obturator internus. As the artery approaches the margin of the ischial ramus, it proceeds above or below the perineal membrane, along the medial margin of the inferior pubic ramus, *en route* to its target structures.

In the male, the internal pudendal artery distal to the perineal artery gives a branch to the bulb of the penis before it divides into the cavernous (deep, cavernosal) and dorsal arteries of the penis (see Fig. 73.12). Given its distribution, the internal pudendal artery distal to its perineal branch has been named the artery of the penis. The artery to the bulb supplies the corpus spongiosum, and the cavernous artery of the penis supplies the corpora cavernosa on each side. The dorsal artery runs on the dorsal aspect of the penis and supplies circumflex branches to the corpora cavernosa and corpus spongiosum that end by anastomosing in the coronal sulcus and supplying the glans penis and its overlying skin.

In the female, a similar branch of the pudendal artery is distributed to the erectile tissue of the corpus spongiosum and vagina. The cavernous artery supplies the corpora cavernosa of the clitoris; the dorsal artery supplies the glans and prepuce of the clitoris.

Branches of the internal pudendal artery are sometimes derived from an accessory pudendal artery, which is usually a branch of the pudendal artery before its exit from the pelvis; effectively, the artery is double before it leaves the pelvis.

Inferior rectal artery

The inferior rectal artery arises just after the internal pudendal artery enters the pudendal canal on the lateral wall of the ischio-anal fossa. It

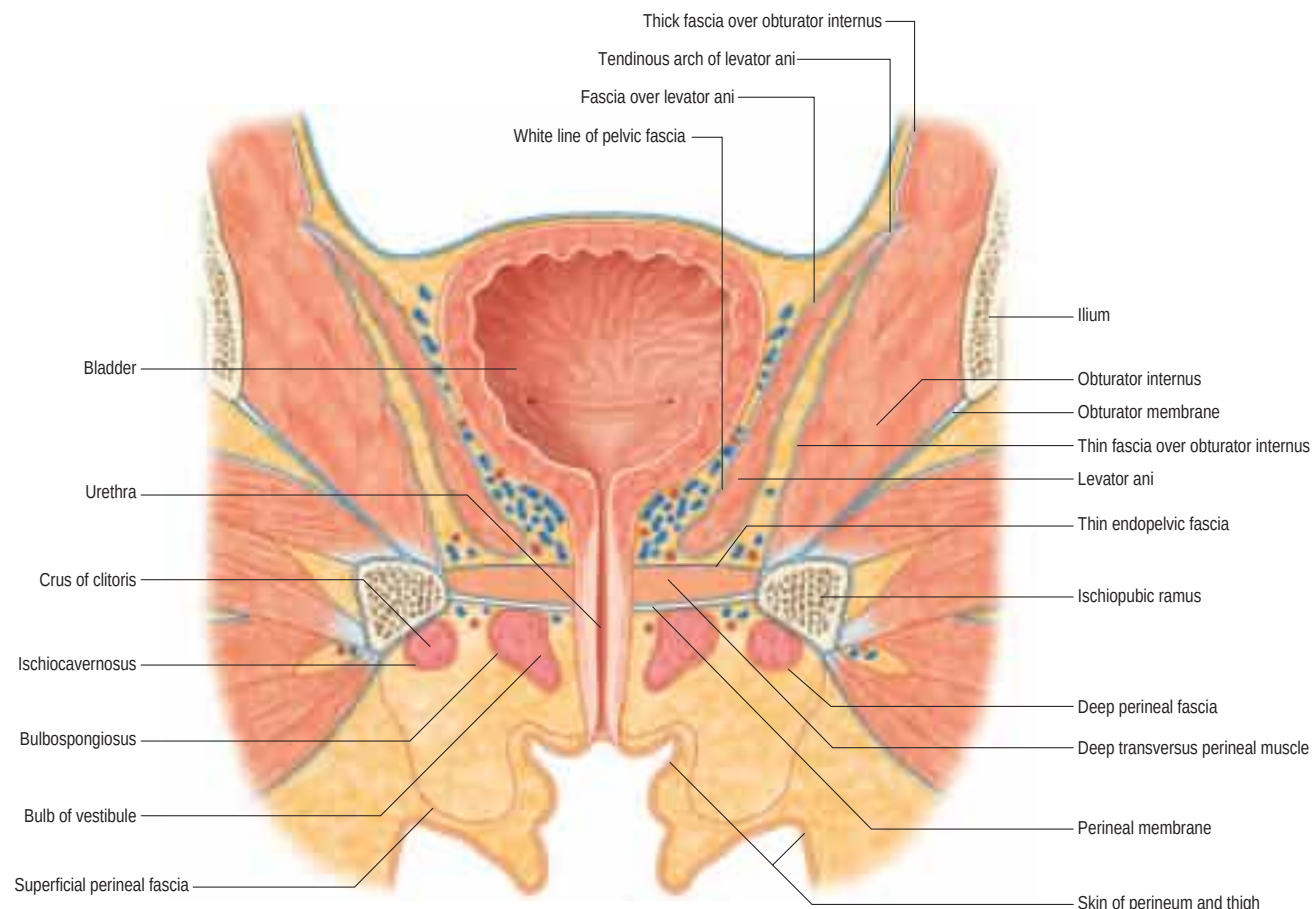


Fig. 73.15 Muscles and fasciae of the female perineum – coronal view. The section passes through the bulb of the clitoris at the level of the urethra. The deep perineal space is continuous with the ischio-anal fossa posteriorly. The visceral and parietal fasciae have been omitted for clarity. The pelvic fascia over the ‘pelvic’ aspect of the deep transverse perinei is very thin and does not form a distinct layer: in places it blends with the parietal pelvic fascia over the inferior aspect of levator ani.

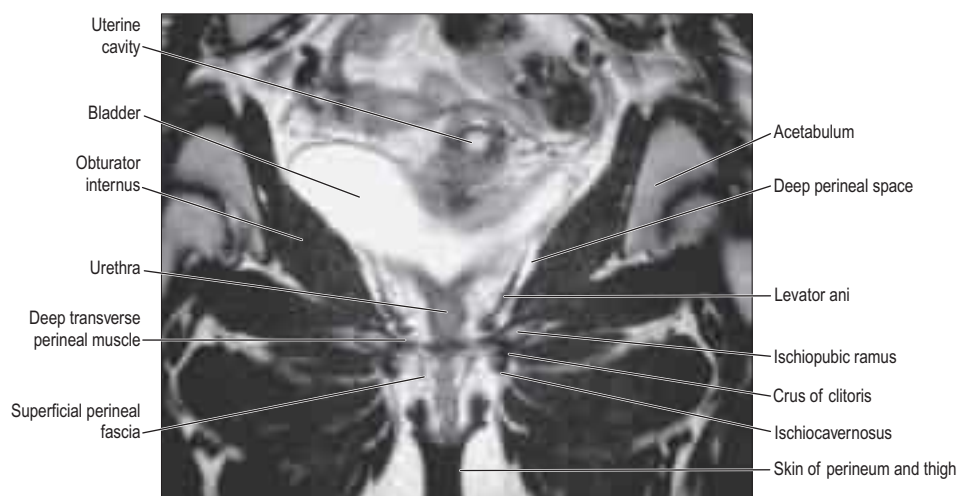


Fig. 73.16 Muscles and fasciae of the female perineum: coronal T2-weighted magnetic resonance image. (Courtesy of Dr J Lee and Ms K Wimpey, Chelsea and Westminster Hospital, London.)

runs anteromedially through the adipose tissue of the ischio-anal fossa to reach the deep portion of the external anal sphincter, and often branches before reaching the sphincter. During dissections of the anal canal, particularly during perineal excisions of the anorectum, the inferior rectal vessels are encountered in the ischio-anal fossa and must be secured before division; otherwise, they tend to retract laterally to the canal, where they can cause troublesome bleeding.

Perineal artery

The perineal artery is a branch of the internal pudendal artery that arises near the anterior end of the pudendal canal and runs through the perineal membrane. In the male, it approaches the scrotum in the superficial perineal space, between bulbospongiosus and ischiocavernosus (see Fig. 76.24). A small transverse branch passes medially, inferior to the superficial transverse perineal muscle, to anastomose with the

contralateral artery and with the posterior scrotal and inferior rectal arteries. It supplies the transverse perinei, the perineal body and the posterior attachment of the bulb of the penis. The posterior scrotal arteries are usually terminal branches of the perineal artery but may also arise from its transverse branch. They are distributed to the scrotal skin and dartos muscle in the male and supply the perineal muscles. In the female, the perineal artery runs an almost identical course to that in the male and gives off the posterior labial arteries (see Fig. 77.4).

Veins of the perineum: internal pudendal veins

The internal pudendal veins are venae comitantes of the internal pudendal artery and unite as a single vessel ending in the internal iliac vein.

The perineal tributaries receive veins from the penile bulb and the scrotum (males), or clitoris and labia (females), and the inferior rectal veins join towards the posterior end of the pudendal canal.

Lymphatic drainage of the perineum

The lymphatics from the skin of the penis and scrotum (male), or skin of the clitoris and labia (female), drain together with lymphatics from the perineal skin to the superficial inguinal nodes and, from there, to the deep inguinal nodes. The glans, corpora cavernosa and corpus spongiosum of the penis or clitoris drain directly to the deep inguinal nodes (see Fig. 78.10).

INNERVATION OF THE PERINEUM: PUDENDAL NERVE (IN THE PERINEUM)

The pudendal nerve gives rise to the inferior rectal and perineal nerves and to the dorsal nerves of the penis or clitoris. The course of its branches parallels the pudendal vessels (see Figs 76.24, 77.4). The pudendal nerve occupies a very constant position over the ischial spine and is readily found. It may be 'blocked' by infiltration with a local anaesthetic applied via a needle passed through the lateral wall of the vagina to numb the perineal and anal skin. It may also be palpated over the ischial spine through the lateral wall of the rectum and motor terminal latencies may be measured.

Inferior rectal nerve

The inferior rectal nerve runs through the medial wall of the pudendal canal with the inferior rectal vessels. It crosses the ischio-anal fossa to supply the external anal sphincter, the lining of the lower part of the anal canal and the circumanal skin. It frequently breaks into terminal branches before reaching the lateral border of the sphincter. Its cutaneous branches that are distributed around the anus overlap the cutaneous branches of the perineal branch of the posterior femoral cutaneous nerve and of the scrotal or labial nerves. The inferior rectal nerve occasionally arises directly from the sacral plexus and crosses the sacrospinous ligament or reconnects with the pudendal nerve. In females, the inferior rectal nerve may supply sensory branches to the lower part of the vagina.

Perineal nerve

The perineal nerve is the inferior and larger terminal branch of the pudendal nerve in the pudendal canal. It runs forwards below the internal pudendal artery and accompanies the perineal artery, dividing into posterior scrotal or labial and muscular branches. The posterior scrotal or labial nerves are usually double; they have medial and lateral branches that run over the perineal membrane and pass forwards in the lateral part of the urogenital triangle with the scrotal or labial branches of the perineal artery. They supply the skin of the scrotum or labia majora, overlapping the distribution of the perineal branch of the posterior femoral cutaneous and inferior rectal nerves. In females, the posterior labial branches also supply sensory fibres to the skin of the lower vagina.

Muscular branches arise directly from the pudendal nerve to supply the superficial transverse perineal muscles, bulbospongiosus, ischiocavernosus, sphincter urethrae and the anterior parts of the external anal sphincter and levator ani. In males, a nerve to the bulb of the urethra leaves the nerve to the bulbospongiosus, pierces this muscle to supply the corpus spongiosum penis, and ends in the urethral mucosa.

Dorsal nerve of the penis or clitoris

The dorsal nerve of the penis or clitoris runs anteriorly above the internal pudendal artery along the ischiopubic ramus, deep to the perineal membrane. It supplies the corpus cavernosum and accompanies the dorsal artery of the penis or clitoris between the layers of the suspensory ligament. In males, the dorsal nerve of the penis runs on the dorsum of the penis to end in the glans.



Bonus e-book images

Fig. 73.7 The relationship between anteroposterior diameters of the right common, internal and external iliac arteries and body surface area (BSA) in boys and girls between the ages of 1 and 16 years.

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